

Reviewed Preprint

v1 • May 30, 2025

Not revised

Reviewed Preprint

v2 • December 23, 2025

Revised by authors

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Competing interests:

No competing interests declared

Funding:

See [page 29](#)

Reviewing editor:

Prayag Murawala, Mount Desert Island Biological Laboratory, United States

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Dorsoventral-mediated *Shh* induction is required for axolotl limb regeneration

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eLife Assessment

This **fundamental** work by Yamamoto and colleagues advances our understanding of how positional information is coordinated between axes during limb outgrowth and patterning. They provide **convincing** evidence that the dorsal-ventral axis feeds into anterior-posterior signaling, and identify the responsible molecules by combining transplantations with molecular manipulations. This work will be of broad interest to regeneration, tissue engineering, and evolutionary biologists.

<https://doi.org/10.7554/eLife.106917.2.sa4>

Abstract

Axolotls (*Ambystoma mexicanum*) exhibit a remarkable ability to regenerate limbs after amputation. Classical experiments have suggested that contact between cells derived from distinct orientations—dorsal, ventral, anterior, and posterior—within the regenerating blastema is necessary for accurate limb pattern formation. However, the molecular basis for this requirement has remained largely unknown. Here, we demonstrate that both dorsal and ventral tissues are required for limb formation via induction of *Shh* expression, which plays a crucial role in limb patterning. Using the accessory limb model (ALM), we induced position-specific blastemas lacking cells derived from a single orientation (anterior, posterior, dorsal, or ventral). Patterned limb formation occurred only in blastemas containing both dorsal- and ventral-derived cells. We further observed that *Shh* expression requires dorsoventral contact within a blastema, highlighting the necessity of dorsoventral contact for inducing *Shh* expression. In addition, we identified WNT10B and FGF2 as dorsal- and ventral-mediated signals, respectively, that create the inductive environment for *Shh* expression. Our findings clarify the role of dorsal and ventral cells in inducing *Shh*, a mechanism that has rarely been studied in the context of limb regeneration and pattern formation. This model provides new insights into how cells with different positional identities drive the regeneration process.

Introduction

Axolotls (*Ambystoma mexicanum*) possess remarkable regenerative abilities, enabling the regeneration of entire limbs after amputation. Among tetrapods (four-limbed vertebrates), only urodele amphibians retain the lifelong ability to regenerate fully developed limbs. Investigating the molecular mechanisms underlying axolotl limb regeneration could provide valuable insights for advancing regenerative medicine in humans, potentially leading to new therapies for tissue repair and organ regeneration.

The limb regeneration process can be divided into two stages: the blastema (regenerating limb primordium) induction process and the limb patterning process. The induction of a blastema, which contains highly proliferative multipotent and unipotent cells, depends on the presence of nerves at the injured region; when a denervated limb is amputated, a blastema is not induced

(Todd, 1823 [↗](#)). This blastema induction process is followed by the limb patterning process. The limb patterning process has been considered to be mainly a recapitulation of limb development. Actually, activation of many developmental genes can be found during this phase. Such sequential events accomplish axolotl limb regeneration.

To accomplish limb patterning, cells need to know its address within a limb and exert its position specific role. Such address of cells has been referred as positional identities. In limb development/regeneration, positional identities are thought to be established along three-dimensional axes—anteroposterior, dorsoventral, and proximodistal. In amniote limb development, WNT7A, secreted from dorsal ectoderm, induces *Lmx1b* expression in the underlying mesenchyme, thereby specifying dorsal identity (Riddle et al., 1995 [↗](#); Vogel et al., 1995 [↗](#); Cygan et al., 1997 [↗](#); Chen et al., 1998 [↗](#); Chen, 2002 [↗](#)). On the ventral side, the ventral identity is specified indirectly by *En1*, which is expressed in the ventral ectoderm, restricts *Wnt7a* expression to the dorsal ectoderm and thereby prevents induction of *Lmx1b* in the ventral mesenchyme (Loomis et al., 1996 [↗](#); Logan et al., 1997 [↗](#); Chen and Johnson, 2002 [↗](#)). Similarly, SHH is known to regulate anteroposterior patterning (Riddle et al., 1993 [↗](#)). In axolotl limb regeneration, the establishment of the dorsoventral positional identities is still largely unknown, except for dorsal-specific *Lmx1b* expression (Shimokawa et al., 2013 [↗](#)). Regarding the anteroposterior axis, *Shh* expression is highly conserved and restricted to the posterior margin. Though the process of the establishment of the positional identities is still veiled in axolotl limb regeneration, it is apparent that the regeneration blastema possesses regional specificity.

For successful limb patterning in limb regeneration, interactions between anterior, posterior, dorsal, and ventral cells within a blastema have been considered essential. The importance of these interpositional interactions was investigated in creating double-half limbs. Amputating the double-half limbs can provide a blastema in a “one-positional-identity-missing” state. For example, when a double-dorsal limb—a chimeric limb surgically generated by excising the ventral half and replacing it with a dorsal half from the contralateral limb while preserving anteroposterior orientation—is amputated, the resulting blastema lacks ventral positional identity. Such blastemas lacking one positional identity can form hypomorphic, spike-like structures or fail to regenerate; in other cases, they regenerate limbs with complete anteroposterior axes accompanied by symmetric dorsoventral duplications (Bryant, 1976 [↗](#); Bryant and Baca, 1978 [↗](#); Burton et al., 1986 [↗](#)). It is noteworthy that, in the non-regenerating cases, structural patterns along the anteroposterior axis appear to be lost even though both anterior and posterior cells should, in principle, be present in a blastema induced from a double-half limb. These observations are consistent with the idea that, during axolotl limb regeneration, signals mediated by dorsal cells must act on ventral cells, and signals mediated by ventral cells must act on dorsal cells, during the limb patterning process. In contrast, in amniote limb development, *Wnt7a/Lmx1b* or *En1* mutants show that limbs can exhibit anteroposterior patterning even when tissues are dorsalized or ventralized—i.e., in the relative absence of ventral or dorsal cells, respectively (Riddle et al., 1995 [↗](#); Chen et al., 1998 [↗](#); Loomis et al., 1996 [↗](#)). Taken together, these findings indicate that the mechanism underlying limb patterning during axolotl regeneration differs from that operating during amniote limb development. Additional experiments involving ectopic contact between opposite regions (e.g., anterior-posterior or dorsal-ventral) of a blastema further support the critical role of interpositional interactions in the limb patterning process, as such interactions can lead to ectopic limb formation (Iten and Bryant, 1975 [↗](#); Bryant and Iten, 1976 [↗](#); Maden, 1980 [↗](#); Stocum, 1982 [↗](#)). These findings suggest that signals mediated by each positional identity are required for successful limb patterning, such that the absence of any one of them leads to failure. To further understand limb regeneration, the molecular basis of these positional identity-mediated signals in limb patterning process should be elucidated.

The accessory limb model (ALM), a non-amputation experimental system, provides a valuable approach for studying axolotl limb regeneration, particularly the role of interpositional interactions (Endo et al., 2015 [↗](#)). In this model, a blastema is induced through skin wounding combined with nerve deviation (Endo et al., 2004 [↗](#); Satoh et al., 2007 [↗](#)). When ALM surgery is performed on the anterior side, the induced blastema (ALM blastema) lacks posterior-derived cells

and is unable to form a properly patterned limb. This limitation can be overcome by supplying posterior-derived cells via a skin graft from the posterior side, enabling the successful regeneration of a patterned limb. Similarly, an ALM surgery can be performed on the posterior, dorsal, or ventral side of a limb; in each case, the induced ALM blastema is expected to lack cells from the corresponding opposite orientation (anterior, ventral, or dorsal, respectively). This condition allows the investigation of blastema induction and limb patterning by selectively removing cells from specific orientations of the limb. Analyzing the characteristics and developmental limitations of these ALM blastemas provides valuable insights into the positional identity-mediated signals that regulate axolotl limb regeneration.

The molecular basis of the interpositional interactions within a blastema in the limb patterning process has been partially elucidated (Nacu et al., 2016 [DOI](#)). In a normal blastema, *Fgf8* and *Shh* are expressed at the anterior and the posterior regions, respectively. A previous study demonstrated that FGF8 and SHH proteins can substitute for anterior and posterior tissues, respectively, to form a patterned limb. Moreover, these proteins function as mutually inductive signals, maintaining each other's expression. These findings suggest that FGF8 and SHH serve as the anterior- and posterior-mediated signals, respectively. In contrast, the molecular basis of dorsal- and ventral-mediated signals remains poorly understood. Previous studies have shown that retinoic acid supplementation can convert the dorsal identity to the ventral, enabling patterned limb formation even in the absence of ventral tissues (McCusker et al., 2014 [DOI](#); Ludolph et al 1990 [DOI](#); Vieira et al., 2019 [DOI](#)). However, the ventral-mediated signal remains uncertain, as retinoic acid induces ventral positional identities rather than acting as the ventral-mediated signal itself, suggesting the existence of another, downstream ventral-mediated signal. Consequently, the mechanisms underlying dorsal- and ventral-mediated signals, as well as how they relate to the anterior- and posterior-mediated signals (FGF8 and SHH, respectively), remain unclear.

In the present study, we investigate the roles of dorsal- and ventral-mediated signals in limb patterning. We confirmed the necessity of dorsoventral tissue contact for limb patterning using the ALM. We found that the induction of *Shh* expression was dependent on the co-existence of dorsal and ventral cells in axolotl limb regeneration. Furthermore, we identified WNT10B and FGF2 as dorsal- and ventral-mediated signals, respectively, by RNA-seq analysis, and confirmed that these factors mediate *Shh* expression in axolotl blastemas. Our results suggest that the crucial role of the dorsal- and ventral-mediated signals in the limb patterning process is to induce *Shh* expression, thereby enabling anteroposterior interaction. These results contribute to understanding how the integration of four positional identities—dorsal, ventral, anterior, and posterior—drives proper limb patterning during axolotl limb regeneration.

Results

Dorsoventral tissue contact is required for limb regeneration in the ALM

We performed the ALM experiment on *Ambystoma mexicanum* to directly test whether dorsoventral tissue contact, as well as anteroposterior tissue contact, is required for limb patterning (Fig. 1 [DOI](#)). The skin was removed from one side (anterior, posterior, dorsal, or ventral) of a limb, and the large nerve fibers running along the center of a limb (*Nervus medianus* and *Nervus ulnaris*, Fig. 1A [DOI](#)) were dissected and rerouted to the wounded region. We supplied the cells from the contralateral side of a limb as a skin graft in experimental conditions to ensure the presence of sufficient cells for patterned limb formation. We conducted eight types of ALM blastema inductions, which can be categorized into two groups: basic ALM blastemas and skin-grafted ALM blastemas. The basic ALM blastemas included the anteriorly induced ALM blastema (AntBL), the posteriorly induced ALM blastema (PostBL), the dorsally induced ALM blastema (DorBL), and the ventrally induced ALM blastema (VentBL), all of which should lack cells from the contralateral side. The skin-grafted ALM blastemas included AntBL with posterior skin grafting (AntBL+P), PostBL with anterior skin grafting (PostBL+A), DorBL with ventral skin grafting (DorBL+V), and VentBL with dorsal skin grafting (VentBL+D); these ALM blastemas should contain

cells with all anteroposterior and dorsoventral orientations. For the ALM experiment, we defined the two large blood vessels running on the anterior and posterior sides at the stylopod level (*Vena cephalica* and *Vena basilica*, Fig. 1A) as the anterior and posterior dorsoventral borders, respectively. At 10 days post-surgery (dps), an ALM blastema was observed in all experimental groups (Fig. 1B–I). Consistent with previous studies (Endo et al., 2004; Nacu et al., 2016), limbs with multiple digits were formed from AntBL+P and PostBL+A (Table 1, Fig. 1F, G), whereas AntBL and PostBL either regressed or formed bump structures (Table 1, Fig. 1B, C). Similarly, DorBL+V and VentBL+D formed limbs (Table 1, Fig. 1H, I), whereas most DorBL and VentBL either regressed or formed bump structures (Table 1, Fig. 1D, E). Notably, only 1 out of 12 DorBL formed a limb, which may have resulted from ventral tissue contamination, as the rerouted nerves originally ran along the ventral side of the humerus. In contrast, none of the 22 VentBL formed multi-digit limbs. These results suggest that both dorsal and ventral tissues, as well as anterior and posterior tissues, are required for successful limb formation.

Gene expression patterns in ALM blastemas

To elucidate the characteristics of the ALM blastemas that typically fail to form limbs (AntBL, PostBL, DorBL, and VentBL), we analyzed gene expression patterns at 10 dps (Fig. 2, Fig. S1). *Lmx1b*, a gene necessary and sufficient for establishing the dorsal identity in mesenchymal cells in developing limb buds (Vogel et al., 1995; Chen et al., 1998; Chen, 2002), was used as a dorsal marker because we previously reported that *Lmx1b* is exclusively expressed in dorsal-derived cells and can be activated in the ALM experiment (Iwata et al., 2020; Yamamoto et al., 2022). The expression patterns of *Fgf8*, *Shh*, and *Lmx1b* were investigated using *in situ* hybridization (ISH). In AntBL and PostBL, *Lmx1b* expression was restricted to the dorsal half of the blastemas (Fig. 2B, G, Fig. S1A, D, $n=5/5$ for both), suggesting the presence of both dorsal and ventral tissues. These expression patterns correspond to the anatomically defined dorsoventral borders (Fig. 1A). In contrast, *Lmx1b* was expressed throughout the entire region of DorBL (Fig. 2L, Fig. S1G, $n=6/6$), whereas no *Lmx1b* expression was detected in VentBL (Fig. 2Q, Fig. S1J, $n=6/6$), suggesting the absence of ventral tissue in DorBL and dorsal tissue in VentBL. Consistent with previous studies, *Fgf8* expression was observed in AntBL (Fig. 2C, Fig. S1B, $n=4/5$), but not in PostBL (Fig. 2H, Fig. S1E, $n=5/5$; Nacu et al., 2016). Conversely, *Shh* expression was detected in PostBL (Fig. 2I, Fig. S1F, $n=5/5$), but not in AntBL (Fig. 2D, Fig. S1C, $n=5/5$). In DorBL and VentBL, *Shh* expression was largely absent (Fig. 2N, Fig. S1I, $n=5/6$; Fig. 2S, Fig. S1L, $n=6/6$), whereas *Fgf8* expression was present in both (Fig. 2M, Fig. S1H, $n=4/6$; Fig. 2R, Fig. S1K, $n=4/6$). In DorBL, *Shh* expression was observed in only 1 out of 6 samples, possibly due to ventral tissue contamination, as described above. These results suggest that *Fgf8* expression is independent of the co-existence of dorsal and ventral tissues, whereas *Shh* expression appears to require their presence.

Induction of *Shh* expression requires both dorsal and ventral cells

The absence of *Shh* expression in most DorBL and VentBL, combined with its consistent expression in all PostBL—which contain both *Lmx1b*-positive and *Lmx1b*-negative regions—suggests that the induction of *Shh* expression depends on the presence of both dorsal and ventral cells (Fig. 2). To test this hypothesis, we performed cell-tracing experiments using GFP-expressing skin from transgenic animals (Fig. 3). We induced VentBL in leucistic animals and then grafted the posterior half of the dorsal skin (VentBL+PD_{gfp}) or the posterior half of the ventral skin (VentBL+PV_{gfp}) from GFP transgenic animals (Fig. 3A). Similarly, DorBL was induced in wild-type animals, and the posterior half of the dorsal skin (DorBL+PD_{gfp}) or the posterior half of the ventral skin (DorBL+PV_{gfp}) from GFP transgenic animals was grafted (Fig. 3A). GFP-positive mesenchymal cells derived from the posterior skin were expected to express *Shh* if provided with a suitable environment, because posteriorly derived cells, not anteriorly derived cells are known to have the competency to express *Shh* in a blastema—i.e., whether a cell is capable of expressing *Shh* depends on its original positional identity (Iwata et al., 2020), but whether it actually expresses *Shh* should depend on the environment in which the cell is placed. If the induction of *Shh* expression depends on the co-existence of dorsal and ventral cells, *Shh* expression in the GFP-

Figure 1. ALM experiments at the four orientations

(A) Schematic image of anatomy at the stylopod level of axolotl limb. HE staining (bright field) and acetylated alpha tubulin, visualized by immunofluorescence (green, dark field), are shown in the right panels. Black and white arrows, respectively, indicate major blood vessels and nerves. H: *humerus*, AHL: *Anconaeus humeralis lateralis*, HAB: *Humeroantebrachialis*, CBL: *Coracobrachialis longus*. (B–I) Blastemas induced at the anterior, posterior, dorsal, or ventral region by skin wounding plus nerve deviation without (B–E) or with (F–I) skin grafting from the opposite side of the limb. (F–I) Limb patterning was observed (n=8/9 for F, 4/7 for G, 7/14 for H, and 7/11 for I, see Table 1 for more detail). Images were captured at 10 and 60 dps. Scale bar = 3 mm.

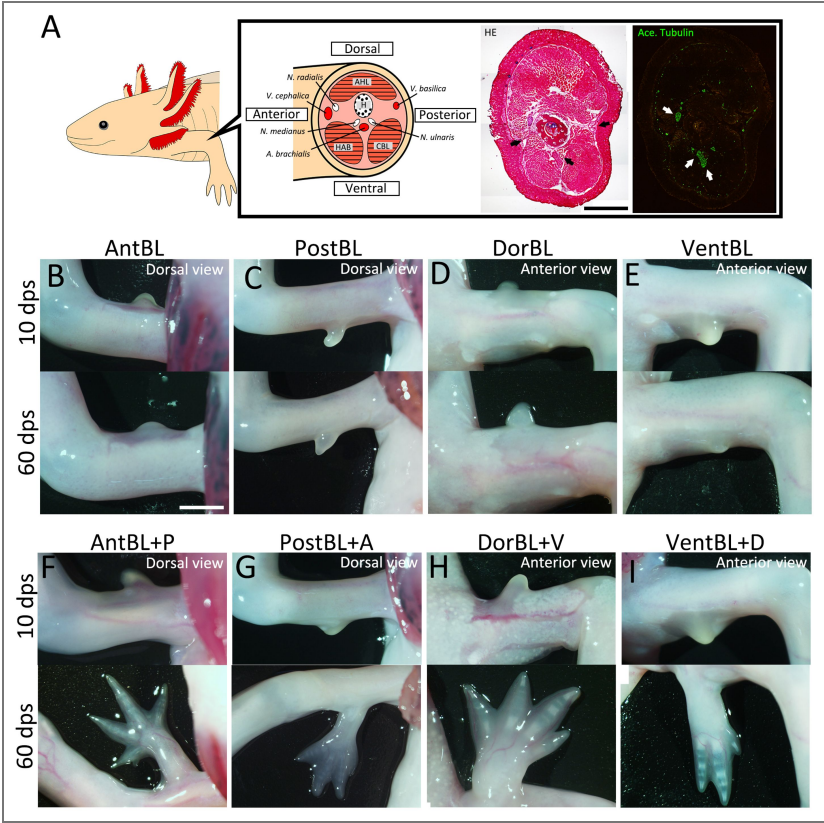


Table 1. The induction rate of bump/limb formation in ALM

	<i>N</i>	Regress		Bump		Limb	
AntBL	8	3	(37.5 %)	5	(62.5 %)	0	(0.0 %)
AntBL+P	9	0	(0.0 %)	1	(11.1 %)	8	(88.9 %)
PostBL	8	6	(75.0 %)	2	(25.0 %)	0	(0.0 %)
PostBL+A	7	2	(28.6 %)	1	(14.3 %)	4	(57.1 %)
DorBL	12	8	(66.7 %)	3	(25.0 %)	1	(8.3 %)
DorBL+V	14	5	(35.7 %)	2	(14.3 %)	7	(50.0 %)
VentBL	22	15	(68.2 %)	7	(31.8 %)	0	(0.0 %)
VentBL+D	11	2	(18.2 %)	2	(18.2 %)	7	(63.6 %)

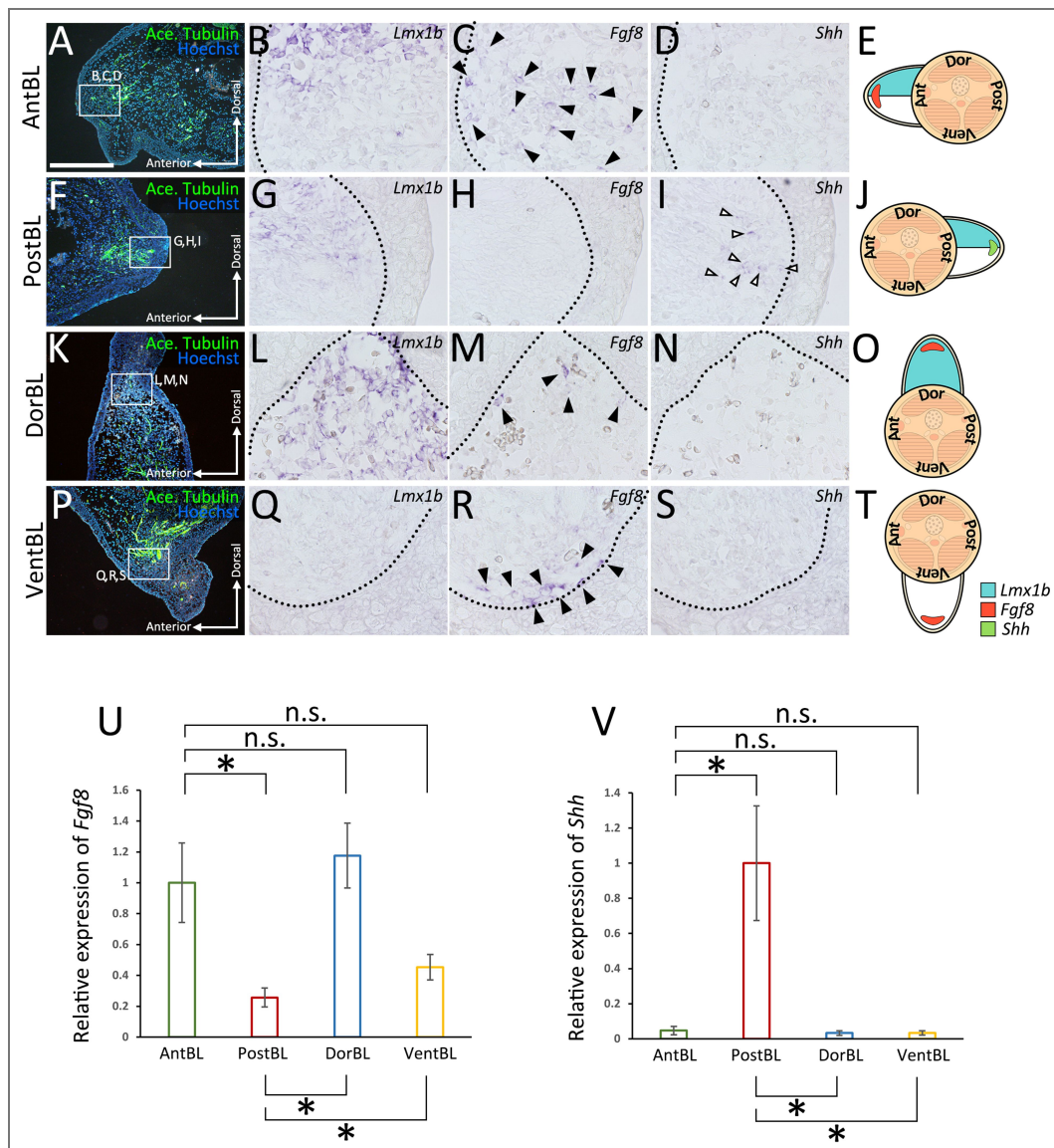


Figure 2. Gene expression patterns of the ALM-induced blastemas

Sections of anteriorly (A–D), posteriorly (F–I), dorsally (K–N), or ventrally (P–S) induced blastemas at 10 dps. Acetylated alpha tubulin (A–P) was visualized by immunofluorescence. Expression of *Lmx1b* (B–Q), *Fgf8* (C–R), and *Shh* (D–S) in the regions indicated by white boxes in (A–P) was visualized by *in situ* hybridization. Images of the entire blastema are provided in Fig. S1. *Fgf8* expression was observed in AntBL (C), DorBL (M), and VentBL (R) (n=4/5, 4/6, and 4/6, respectively). *Shh* expression was observed in PostBL (I) (n=5/5). In each case, these expression patterns of *Fgf8* and *Shh* were focal and only a few cells expressed *Fgf8* or *Shh*. Black and white arrowheads indicate the signals of *Fgf8* and *Shh* expression, respectively. The dotted line indicates the epithelial-mesenchyme border. (E–T) Schematic images of gene expression patterns. (U, V) Quantitative analysis of *Fgf8* and *Shh* expression in ALM blastemas (n=5 for all groups). n.s.; no significant difference, * $p < 0.05$, ** $p < 0.005$ (two-tailed Welch's *t*-test). Scale bar in (A) = 700 μ m.

positive cells should be observed in VentBL+PD_{gfp} and DorBL+PV_{gfp} but not in DorBL+PD_{gfp} or VentBL+PV_{gfp}. Samples were collected at 10 dps, and the expression of *Shh* and *Lmx1b* was analyzed using ISH. In all four experimental groups, *Lmx1b* expression patterns in the GFP-negative regions were consistent with those in DorBL and VentBL (Fig. 3D, F, H, J). In DorBL+PD_{gfp} and VentBL+PD_{gfp}, *Lmx1b* expression was observed in GFP-positive mesenchymal cells, whereas GFP-positive cells in DorBL+PV_{gfp} and VentBL+PV_{gfp} were *Lmx1b*-negative (Fig. 3C–F). These *Lmx1b* expression patterns suggest that dorsoventral tissue contact was present in VentBL+PD_{gfp} and DorBL+PV_{gfp} but absent in DorBL+PD_{gfp} and VentBL+PV_{gfp}. In the GFP-positive cells derived from PD_{gfp} skin, *Shh* expression was observed in VentBL+PD_{gfp} (Fig. 3C, n=4/6) but not in DorBL+PD_{gfp} (Fig. 3E, n=6/6). Similarly, in the GFP-positive cells derived from PV_{gfp} skin, *Shh* expression was observed in DorBL+PV_{gfp} (Fig. 3I, n=3/6) but not in VentBL+PV_{gfp} (Fig. 3G, n=7/7). In addition, *Shh* expression was observed in some GFP-negative mesenchymal cells in samples where *Shh* expression was observed in GFP-positive cells. These results suggest that the induction of *Shh* expression in posteriorly derived cells requires the co-existence of dorsal and ventral cells.

Limb formation in the absence of dorsoventral tissue contact

DorBL and VentBL failed to form limbs (Fig. 1D, E). *Shh* expression was absent in these blastemas, whereas *Fgf8* was expressed (Fig. 2). Results from the cell-tracing experiments suggest that *Shh* expression requires the co-existence of dorsal and ventral cells (Fig. 3). Based on these findings, we hypothesized that while the co-existence of dorsal and ventral cells is necessary to induce *Shh* expression, it is not directly required for limb patterning if SHH protein is externally provided. To test this hypothesis, we overexpressed *Shh* in DorBL and VentBL (Fig. 4A, B). As a positive control, *Shh* was overexpressed in AntBL, where *Fgf8* was expressed but *Shh* expression was absent (Fig. 2C, D), because a previous study has shown that *Shh* overexpression to AntBL can induce limb patterning (Nacu et al., 2016). As a result, *Shh*-electroporated AntBL, DorBL, and VentBL formed limbs (n=10/19, 6/18, and 8/14, respectively, Fig. 4C–G). In contrast, GFP-electroporated DorBL and VentBL (negative controls) failed to form limbs (n=7/7 and 6/6, respectively). We further analyzed the anatomy of the induced limbs (Fig. 4H–K, Fig. S2). Because ALM-induced limbs frequently exhibit abnormal and highly variable morphologies, which makes it difficult to use consistent anatomical landmarks such as particular digits or muscle groups, we focused our analysis on morphological symmetry. We found that limbs formed from DorBL and VentBL exhibited symmetric structures along the dorsoventral axis, suggesting double-dorsal and double-ventral structures, respectively (Fig. 4J, K, Fig. S2C, D). In contrast, limbs formed from AntBL, which contained both *Lmx1b*-positive and *Lmx1b*-negative regions (Fig. 2B), exhibited asymmetric structures similar to those of normal limbs (Fig. 4H, I, Fig. S2A, B). To evaluate the symmetry of the limbs, we applied a machine learning-based method using ilastik because staining intensity varied among samples, such that a region identified as “muscle” in one sample could be assigned differently in another if classification were based solely on color. The machine-learning classifier trained separately for each sample allowed us to group the same tissues consistently within that sample irrespective of intensity differences. To minimize the effects of curvature or fixation-induced distortion, the boxes with a width of 400 μm were selected, and the angle was adjusted so that the outer contour (epidermal surface) was aligned symmetrically; this procedure was applied uniformly across all conditions to avoid bias. Each pixel in the images was classified into five classes (Classes 1 to 5, Fig. 4L). In this process, we annotated regions of background (Class 1), cartilage (Class 2), muscle (Class 3), other connective tissue (Class 4), and epidermis (Class 5) as training data for classification. Using these annotated regions as references, pixels were automatically classified into the five classes. Each class was assumed to primarily represent these tissues and regions with similar characteristics. Symmetry scores were then calculated for each class individually and for the combined set of all classes (Fig. 4M–R, see Materials and Methods). The analysis revealed that symmetry scores for Classes 2 and 3, which encompass cartilage and muscle, and scores for the combined set of all classes were significantly higher in limbs formed from DorBL and VentBL compared to intact limbs (Fig. 4N, O, R). In contrast, the differences in symmetry scores for Classes 1, 4, and 5 in these limbs were

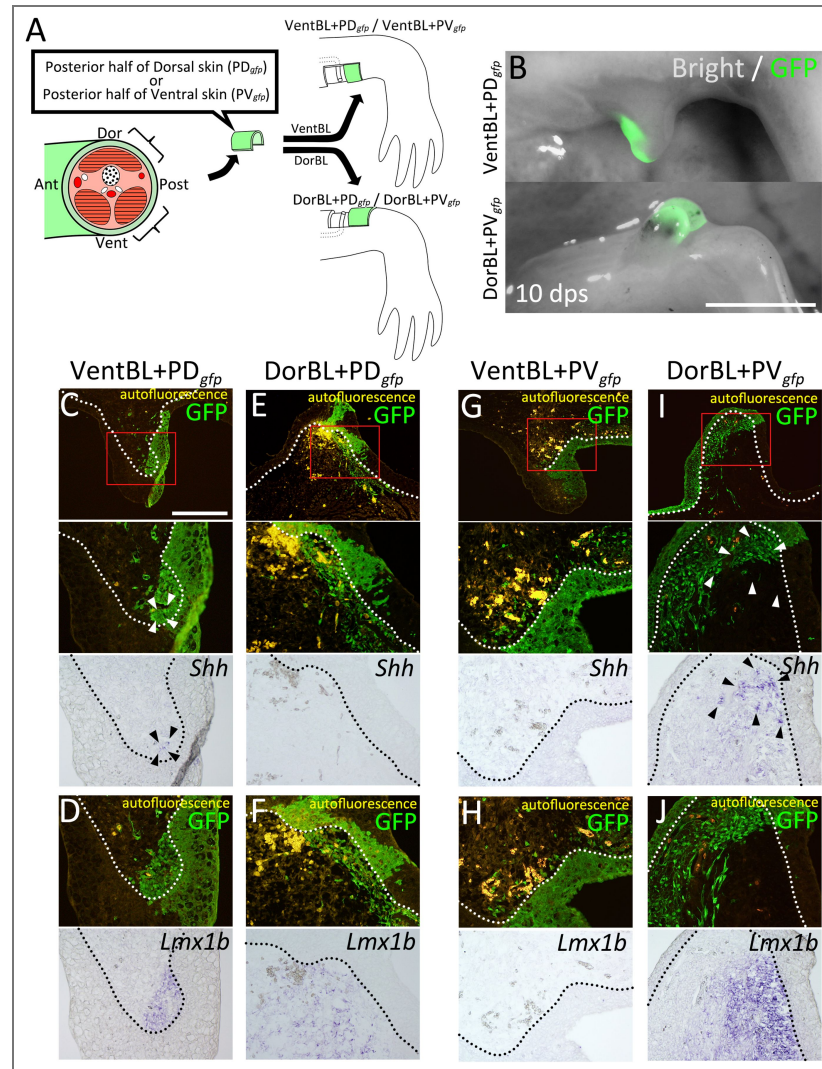


Figure 3. Co-existence of dorsal and ventral cells induces *Shh* expression

(A) Experimental scheme. Posterior half of dorsal (PD_{gfp}) or ventral (PV_{gfp}) GFP-expressing skin was grafted on VentBL (VentBL+PD_{gfp}; C, D / VentBL+PV_{gfp}; G, H), or DorBL (DorBL+PD_{gfp}; E, F / DorBL+PV_{gfp}; I, J) region. (B) Induced blastemas at 10 dps; images of bright and dark fields are merged. (C–J) Dark and bright fields of the same sections of induced blastemas at 10 dps. Red boxes in (C–I) indicate the corresponding regions of lower images. Expression of *Shh* and *Lmx1b* were visualized by *in situ* hybridization. GFP signals were visualized by immunofluorescence. Arrowheads indicate the cells expressing *Shh*. *Shh* expression was observed in VentBL+PD_{gfp} (C) and DorBL+PV_{gfp} (I) (n=4/6 and 3/6, respectively), but not in DorBL+PD_{gfp} (E) and VentBL+PV_{gfp} (G) (n=6/6 and 7/7, respectively). The dotted line indicates the epithelial-mesenchyme border. For all samples, we collected serial sections spanning the entire blastema. For blastemas in which *Shh* expression was observed, we present representative sections showing the signal. For blastemas without detectable *Shh* expression, we present a section from the central region that contains GFP-positive cells. Scale bar = 3 mm (B) and 700 μm (C).

relatively small (Fig. 4M, P, Q [↗](#)), likely because the axis of symmetry was manually set to maximize the external shape as symmetrically as possible. No significant differences in symmetry scores across all classes were observed between intact limbs and limbs formed from AntBL. These results suggest that limbs formed from DorBL and VentBL exhibit symmetric internal structures compared to normal limbs. Therefore, *Shh* overexpression appears to compensate for the lack of co-existence of dorsal and ventral cells in limb patterning, without inducing new dorsal or ventral identities. Thus, we conclude that the co-existence of both dorsal and ventral cells is critical for inducing *Shh* expression, which in turn is essential for limb patterning.

The molecular basis of the dorsal-mediated and ventral-mediated signals

To investigate the molecular basis of dorsal- and ventral-mediated signals, we performed RNA-seq analysis on DorBL and VentBL and identified differentially expressed genes (DEGs) between the two groups ($p < 0.05$, Fig. 5A [↗](#)). Among the DEGs, we specifically focused on genes annotated as “intercellular signaling molecules” in PANTHER and identified 21 genes (Fig. 5B [↗](#)). In this analysis, we found that 5 genes were expressed at higher levels in DorBL and 16 genes were expressed at higher levels in VentBL (Fig. 5B [↗](#)). Notably, *Wnt4*, *Wnt10b*, *Fgf2*, *Fgf7*, and *Tgfb2*, which encode secreted proteins, belong to the WNT, FGF, and TGF β families, which play key roles in major signaling pathways regulating limb developmental processes and we focused on these 5 genes. To examine whether these genes regulate the induction of *Shh* expression in ALM blastemas, we overexpressed each gene in either DorBL or VentBL. As a result, *Shh* expression was observed in *Wnt10b*-electroporated VentBL ($n=4/5$) and *Fgf2*-electroporated DorBL ($n=5/7$, Fig. 5C [↗](#)). In contrast, *Shh* expression was not detected in *Wnt10b*-electroporated DorBL ($n=6/6$) or *Fgf2*-electroporated VentBL ($n=5/5$). Similarly, *Shh* expression was not detected in *Fgf7*- or *Tgfb2*-electroporated DorBL ($n=5/5$ for both) or in *Wnt4*-electroporated VentBL ($n=6/6$). To confirm the ISH data, we performed RT-qPCR on *Fgf2*- or GFP (control)-electroporated DorBL and *Wnt10b*- or GFP (control)-electroporated VentBL and observed significant upregulation in *Shh* expression (Fig. 5D, E [↗](#)). In *Wnt10b*-electroporated VentBL, *Axin2*, a downstream transcriptional target of canonical WNT signaling, and *Lef1*, a canonical WNT pathway effector expressed in axolotl limb mesenchyme (Glötzer et al., 2022 [↗](#)), were also upregulated (Fig. 5F, G [↗](#)). Next, to confirm *Shh* induction by WNT signaling in VentBL, we treated VentBL with 1 μ M 6-bromoindirubin-3-oxime (GSK-3 Inhibitor, BIO). As a result, *Shh* expression was relatively higher in BIO-treated VentBL compared to DMSO-treated control VentBL (Fig. 5B, D, E [↗](#)). Additionally, symmetric limbs were formed in *Wnt10b*-electroporated VentBL ($n=3/12$), BIO-treated VentBL ($n=3/8$), and in *Fgf2*-electroporated DorBL ($n=5/10$), consistent with the results of *Shh* overexpression (Fig. 5H–J [↗](#), Fig. 5A, C, F–H [↗](#), Fig. 5A, B [↗](#)). These findings suggest that WNT10B, expressed highly in dorsal blastema cells, and FGF2, expressed highly in ventral blastema cells, function as the dorsal- and ventral-mediated signals, respectively, to induce *Shh* expression, which subsequently supports limb patterning.

To test whether our model applies to normal regeneration, we analyzed *Wnt10b* and *Fgf2* expression in amputation-induced blastemas (Fig. 5S [↗](#)). We first performed *in situ* hybridization on blastemas at several stages (early bud [EB], middle bud [MB], and late bud [LB]), but the signals were weak and inconsistent, and we could not reliably detect clear expression domains (Fig. 5S [↗](#)). We then performed RT-qPCR on manually microdissected dorsal and ventral halves of MB blastemas (Fig. 5S [↗](#)). We found that *Wnt10b* and *Fgf2* were expressed at significantly higher levels in the dorsal and ventral halves, respectively, compared to the opposite half. This dorsoventral-biased expression of *Wnt10b* and *Fgf2* is consistent with our RNA-seq data from ALM blastemas. We next quantified *Wnt10b*, *Fgf2*, and *Shh* expression across stages (intact, EB, MB, LB, and early digit [ED]) and found that *Wnt10b* and *Fgf2* expression peaked at the MB stage, whereas *Shh* expression peaked later, at the LB stage (Fig. 5S [↗](#)). This temporal offset in *Shh* upregulation relative to *Wnt10b* and *Fgf2* supports a model in which WNT10B and FGF2 act upstream to induce *Shh* expression.

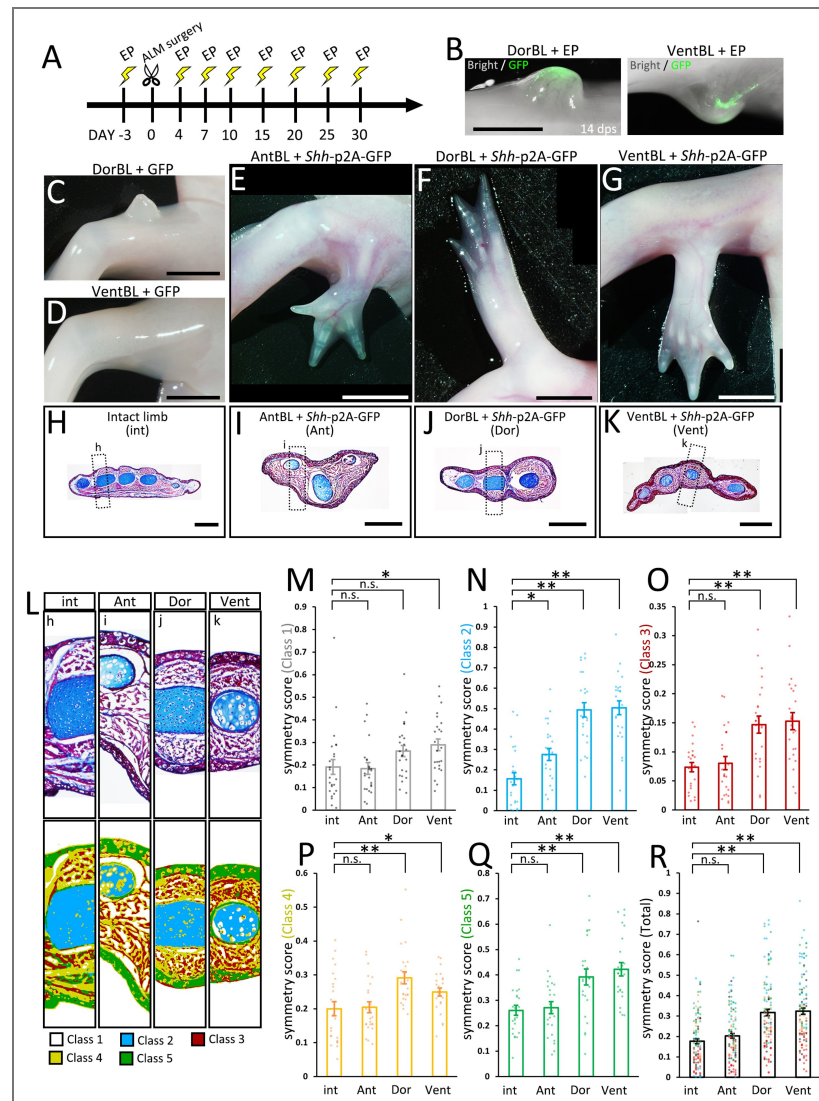


Figure 4. Limb formation without the co-existence of dorsal and ventral cells by Shh overexpresion

(A) Experimental scheme. *Shh*-p2A-GFP- or GFP-containing pCS2 vector was electroporated (EP) into AntBL, DorBL, and VentBL. (B) Induced blastemas at 14 dps; images of bright and dark fields are merged. (C–G) Phenotypes at 90 dps. Limb patterning was observed in (E–G) (n=10/19, 6/18, and 8/14, respectively). (H–K) Histological analysis of intact and induced limbs. Standard Masson’s trichrome staining was performed on the transverse sections. The dotted boxes indicate the regions shown in (L). (L) Upper panels: analyzed regions for calculating symmetry scores. Lower panels: images after pixel classification by machine learning. (M–R) Symmetry scores of each class. Scores obtained from the same limb are plotted at the same x-coordinates. n.s.; no significant difference, * $p < 0.05$, ** $p < 0.005$ (two-tailed Welch’s *t*-test). Scale bar = 2 mm (B), 4 mm (C–G), and 1 mm (H–K).

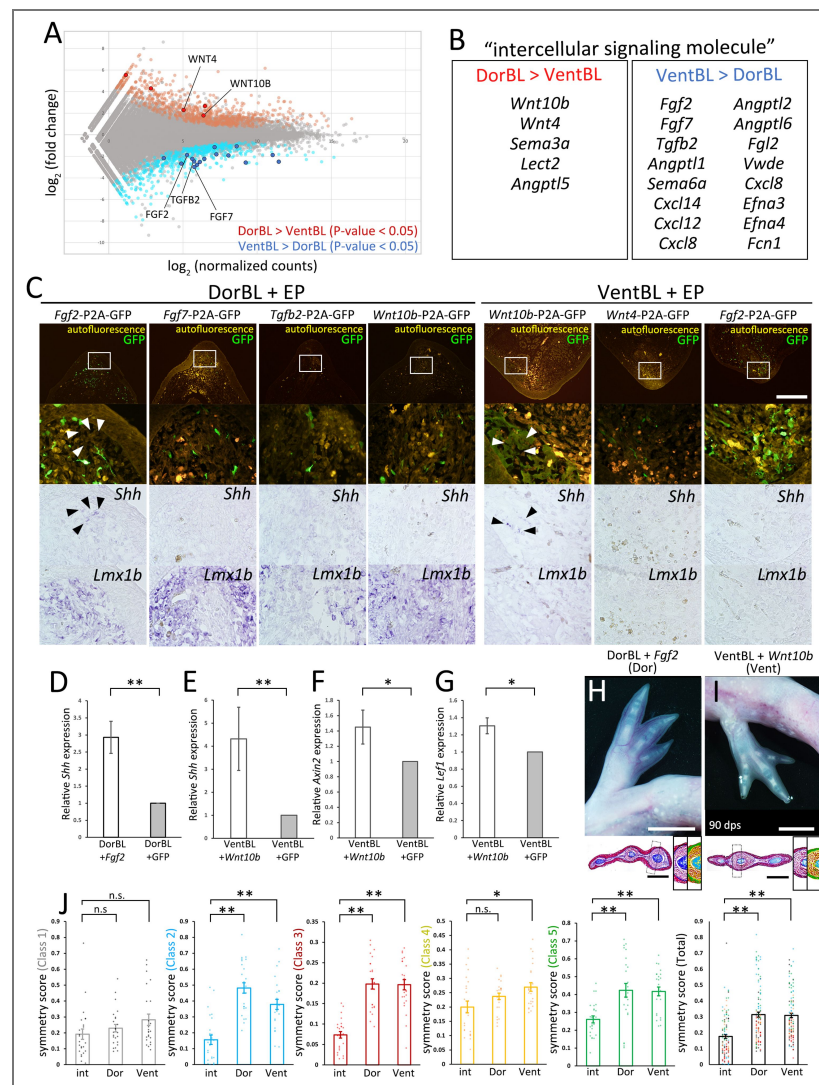


Figure 5. Identification of candidate molecules of the dorsal- and ventral-mediated signals

RNA-Seq was performed on DorBL and VentBL at 10 dps. (A) MA plot of the result. (B) List of DEGs annotated as “intercellular signaling molecules.” (C) Bright and dark fields of sections of DorBL and VentBL at 10 dps with candidate genes introduced. Expression of *Shh* and *Lmx1b* was visualized by *in situ* hybridization, and GFP signals were visualized by immunofluorescence. Arrowheads indicate the cells expressing *Shh*. The white boxes indicate the regions of the lower panels. Images of dark and bright fields of *Shh* are obtained from the same section. For samples with detectable *Shh* expression, the window was placed in the region where the signal was observed, and for conditions without detectable *Shh* expression, the window was positioned in a comparable region containing GFP-positive cells. *Shh* expression was observed in *Wnt10b*-electroporated VentBL (n=4/5) and *Fgf2*-electroporated DorBL (n=5/7), but not in *Wnt10b*-electroporated DorBL (n=6/6), *Fgf2*-electroporated VentBL (n=5/5), *Fgf7*-electroporated DorBL (n=5/5), *Tgfb2*-electroporated DorBL (n=5/5), or in *Wnt4*-electroporated VentBL (n=6/6). (D, E) Quantitative analysis of *Shh* expression in *Fgf2*- or GFP-electroporated DorBL and *Wnt10b*- or GFP-electroporated VentBL (n=7 for both). In each case, *Fgf2* or *Wnt10b* was electroporated into the DorBL or VentBL induced in the left limb, and GFP was electroporated into the contralateral right limb of the same animal. (F, G) Quantitative analysis of *Axin2* and *Lef1* expression in *Wnt10b*- or GFP-electroporated VentBL (n=7 for both). (H, I) The limbs formed from VentBL with *Wnt10b* and DorBL with *Fgf2* at 90 dps. Histological analysis and pixel classification was performed in the same way as in Fig. 4. The dotted boxes indicate the regions shown in the right panels. (J) Symmetry scores of each class. Scores obtained from the same limb are plotted at the same x-coordinates. The plots of “int” are the same plots as in Fig. 4. n.s.; no significant difference, **p*<0.05, ***p*<0.005 (two-tailed paired *t*-test for D–G, and two-tailed Welch’s *t*-test for J). Scale bar = 700 μm (C), 4 mm (H, I, upper panels), 1 mm (H, I, lower panels).

To identify the cell populations expressing *Wnt10b* and *Fgf2* during normal regeneration, we reanalyzed published single-cell RNA-seq data from a 7 dpa (MB) blastema (Li et al., 2021 [Fig. S6](#)). The dataset was reclustered, and clusters were assigned using known markers (*Prrx1* for mesenchyme and *Krt17* for epithelium, [Fig. S6A, B](#)). As expected, *Lmx1b*, *Fgf8*, and *Shh* were detected in the mesenchymal cluster ([Fig. S6C, F, G](#)). *Fgf2* was also expressed in the mesenchymal cluster ([Fig. S6E](#)). In contrast, *Wnt10b* expression was detected in both mesenchymal and epithelial clusters ([Fig. S6D](#)). Both *Wnt10b* and *Fgf2* were expressed in only a few cells, consistent with the *in situ* hybridization data ([Fig. S5A](#)). We then examined the relationships between these genes. *Fgf8* and *Shh* were expressed in both *Lmx1b*-positive and *Lmx1b*-negative cells ([Fig. S6H, I](#)), but *Fgf8* and *Shh* themselves were mutually exclusive ([Fig. S6M](#)). These expression patterns of *Fgf8*, *Shh*, and *Lmx1b* in the normal blastema are consistent with those observed in ALM blastemas ([Fig. 2](#)). For *Wnt10b* and *Fgf2*, their expression did not follow *Lmx1b* expression ([Fig. S6J, K](#)), and *Wnt10b* and *Fgf2* themselves were not exclusive ([Fig. S6L](#)). Together with the RT-qPCR data ([Fig. S5B](#)), these results suggest that *Wnt10b* and *Fgf2* are not exclusively confined to purely dorsal or ventral cells at the single-cell level, even though they show dorsoventral bias when assessed in bulk tissue.

Limb formation without nerve deviation and ventral skin grafting

A previous study demonstrated that a cocktail of BMP2, FGF2, and FGF8 can substitute for nerve deviation in the ALM experiment on the anterior region, suggesting that nerves contribute to blastema induction by supplying these proteins (Makanae et al., 2014 [Fig. 5](#)). We found that FGF2 can also substitute for ventral tissue in DorBL ([Fig. 5](#)), suggesting that FGF2 serves not only as part of the nerve factors in blastema induction but also as the ventral-mediated signal in limb patterning. To investigate whether supplementation with BMP2, FGF2, and FGF8 to the dorsal region could substitute for both nerve deviation and ventral skin grafting, we performed experiments involving gelatin beads soaked in these proteins ([Fig. 6](#)). The dorsal or ventral skin was removed at the zeugopod level, and a BMP2+FGF2+FGF8-soaked gelatin bead was grafted at 3 dps without nerve deviation. Blastema induction was observed in both experimental groups ([Fig. 6A, B](#), upper panels). We investigated gene expression patterns in the induced blastemas and found that both *Fgf8* and *Shh* were expressed in blastemas induced at the dorsal site (n=8/8), whereas *Shh* expression was absent in ventral blastemas (n=5/5), although *Fgf8* expression was detected in most cases (n=4/5, [Fig. 6C, D, F, G](#)). *Lmx1b* expression patterns corresponded to those observed in DorBL and VentBL ([Fig. 6E, H](#), n=8/8 and 5/5, respectively). Furthermore, limb patterning was observed in the dorsal groups (n=12/20) but not in the ventral groups (n=17/17, [Fig. 6A, B](#), lower panels). These findings demonstrate that a straightforward procedure involving dorsal skin wounding and supplementation with BMP2, FGF2, and FGF8 is sufficient to induce accessory limb formation. This method enabled the induction of an ectopic limb without surgical nerve deviation, but a previous study has shown that fine nerve ingrowth can still occur when blastemas are induced by a BMP2+FGF2+FGF8-soaked bead (Makanae et al., 2014 [Fig. 5](#)). These recruited nerves may be functional after blastema induction. Remarkably, this method also enabled the induction of multiple limbs within the same limb. Grafting BMP2+FGF2+FGF8-soaked beads to multiple dorsal sites resulted in the formation of multiple limbs along the proximodistal axis (n=16/35, [Fig. 6I–L](#)). Despite careful implantation designed to avoid injuring deep tissues, one sample displayed a fusion of the stylopod with the host humerus—a phenotype associated with deep wounding (Sato et al., 2010 [Fig. 1](#); Makanae et al., 2014 [Fig. 5](#)). This suggests that contributions from a broader cellular population cannot be excluded. However, because this fusion was not consistently observed, and because ectopic limbs induced at the forearm (zeugopod) level did not exhibit such fusion (n=1/6 for stylopod-level inductions; n=0/10 for zeugopod-level inductions, [Fig. 6L1, L2](#)), the data suggest that most ectopic limbs would have developed without substantial ventral-cell contribution. These results refine our understanding of the role of FGF2 in the ALM, particularly its critical function on the dorsal side.

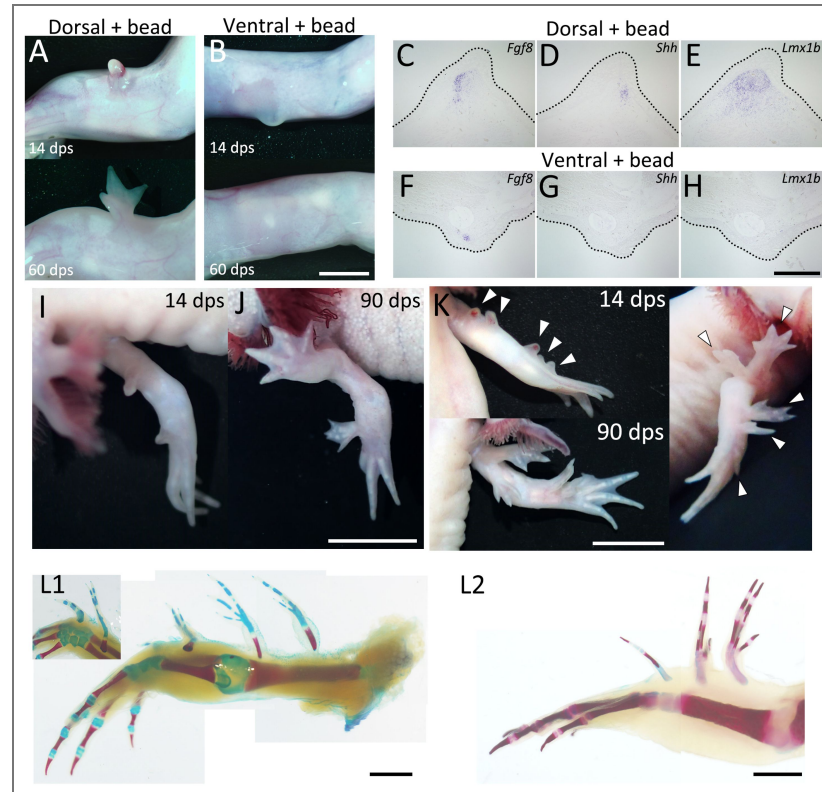


Figure 6. Limb formation at the dorsal region by BMP2+FGF2+FGF8 supplementation

(A, B) 14 and 60 dps phenotypes of BMP2+FGF2+FGF8 supplementation by bead grafting at the dorsal (A) or ventral (B) region. Neither nerve deviation nor skin grafting was performed. Limb patterning was observed in the dorsal group (A) ($n=12/20$) but not in the ventral group (B) ($n=0/17$). (C–H) Expression patterns of *Fgf8*, *Shh* and *Lmx1b* of induced blastemas at 10 dps. Gene expression was visualized by *in situ* hybridization. The dotted line indicates the external shape of the blastema. *Fgf8* expression was detected in both dorsal and ventral groups ($n=8/8$ for C and $5/5$ for F), whereas *Shh* expression was detected only in the dorsal group (D) ($n=8/8$) and not in the ventral group (G) ($n=0/5$). (I–J) Phenotype obtained by BMP2+FGF2+FGF8 supplementation to two dorsal regions of an identical limb. (K, L) Phenotype obtained by BMP2+FGF2+FGF8 supplementation to five dorsal regions of an identical limb ($n=16/35$). Scale bar = 3 mm (B), 700 μ m (H), 2 cm (J), 1 cm (K, L).

Discussion

Dorsal- and ventral-mediated signals are required for the induction of *Shh* expression

In axolotl limb regeneration, cells derived from all four positional origins—anterior, posterior, dorsal, and ventral—are required for an induced blastema to form a limb. This was confirmed through the ALM experiments (Fig. 1 [↗](#)). The gene expression patterns in these ALM blastemas revealed distinct molecular characteristics for each group (Fig. 2 [↗](#)). *Fgf8* expression in AntBL and *Shh* expression in PostBL are consistent with a previous report (Nacu et al., 2016 [↗](#)). The absence of SHH in AntBL and of FGF8 in PostBL likely accounts for their failure to form limbs. *Lmx1b* expression in the dorsal half of both AntBL and PostBL suggests that both dorsal and ventral tissues co-existed in these blastemas (Fig. 2B, G [↗](#), Fig. S1A, D [↗](#)). In contrast, DorBL and VentBL exhibited distinct *Lmx1b* expression patterns, suggesting that most cells in DorBL and VentBL are derived from dorsal or ventral origins, respectively (Fig. 2L, Q [↗](#), Fig. S1G, J [↗](#)). A possible explanation for the lack of *Shh* expression in DorBL and VentBL is that the induction of *Shh* expression may depend on the co-existence of dorsal and ventral cells, although *Fgf8* can be expressed independently of the co-existence of dorsal and ventral cells (Fig. 2M, N, R, S, U, V [↗](#), Fig. S1H, I, K, L [↗](#)). This idea is supported by the cell-tracing experiment (Fig. 3 [↗](#)). In these assays, the posteriorly derived cells expressed *Shh* only when both dorsal and ventral cells were present. These results strongly suggest that the co-existence of dorsal and ventral cells is necessary for inducing *Shh* expression. Consequently, one of the essential roles of the dorsal and ventral cells in limb patterning is to mediate signals required for *Shh* induction in the posterior cells, which facilitates the anteroposterior interaction.

Our findings clarify the hierarchical relationship between dorsal- and ventral-mediated signals and anteroposterior interaction. Our data demonstrate that dorsal and ventral cells are essential for inducing *Shh* expression in a regenerating blastema. Furthermore, we showed that DorBL and VentBL, which typically fail to form a limb, could form patterned limbs with ectopic *Shh* expression, even in the absence of dorsal and ventral cell co-existence (Fig. 4 [↗](#)). Furthermore, such limbs exhibited dorsally or ventrally symmetric structures (Fig. 4F, G, J–R [↗](#), Fig. S2C, D [↗](#)), highlighting that dorsoventral patterning depends on the cell origin. Thus, we concluded that dorsal- and ventral-mediated signals are essential for patterned limb formation via inducing *Shh* expression.

WNT10B and FGF2 as dorsal- and ventral-mediated signals

We identified *Wnt10b* and *Fgf2* as candidate genes encoding the dorsal- and ventral-mediated signals required to induce *Shh* expression, respectively (Fig. 5 [↗](#)). Our DEG analysis between DorBL and VentBL revealed higher *Wnt10b* expression in DorBL and higher *Fgf2* expression in VentBL (Fig. 5A, B [↗](#)). *Wnt10b*-electroporation in VentBL and *Fgf2*-electroporation in DorBL induced *Shh* expression and subsequent limb patterning (Fig. 5C–E [↗](#)). *Wnt4* expression was also elevated in DorBL, while *Fgf7* and *Tgfb2* were upregulated in VentBL. However, we did not observe *Shh* induction in the *Fgf7*-electroporated or *Tgfb2*-electroporated DorBL or in the *Wnt4*-electroporated VentBL. FGF7 is known to be capable of inducing the apical ectodermal ridge (AER) in chick limb development (Yonei-Tamura et al., 1999 [↗](#)), but little is known about its role in limb regeneration. In newt limb regeneration, KGFR, which acts as an FGF7 receptor, is expressed in the basal layer of the wound epithelium, while FGFR1, which acts as an FGF2 receptor, is primarily expressed in the mesenchyme (Poulin et al., 1993 [↗](#)). This differential expression pattern suggests that FGF2 and FGF7 target distinct cell populations, potentially explaining the differences in their ability to induce *Shh* expression. The difference in the ability of *Wnt10b* and *Wnt4* to induce *Shh* expression in VentBL may reflect differences in how these ligands activate downstream WNT signaling programs. WNT10B is a potent activator of the canonical WNT signaling pathway (Bennett et al., 2005 [↗](#)), although WNT10B has also been reported to trigger a β -catenin-independent pathway (Lin et al., 2021 [↗](#)). Similarly, WNT4 can activate both the canonical WNT

signaling pathways and a non-canonical, β -catenin-independent pathway (Li et al., 2013 [↗](#); Li et al., 2019 [↗](#)). We also observed *Shh* induction in BIO-treated VentBL, indicating that the canonical WNT signaling pathway regulates *Shh* expression. However, it is uncertain why *Shh* expression was observed in *Wnt10b*-electroporated VentBL, but not in *Wnt4*-electroporated VentBL. One possible explanation is that different WNT ligands can engage the same receptors (Frizzled/LRP6) yet elicit distinct downstream programs, suggesting that such ligand-specific outputs may vary depending on cell context (Voss et al., 2025 [↗](#)). Regarding *Tgfb2*, TGF- β signaling has been implicated in blastema induction during salamander limb regeneration (Sader and Roy, 2022 [↗](#); Lévesque et al., 2007 [↗](#)). However, little is known about the relationship between TGF- β signaling and *Shh* induction in limb development and regeneration, suggesting that TGF- β signaling does not play a primary role in *Shh* induction. Consistent with this, we did not observe *Shh* induction in the *Tgfb2*-electroporated DorBL, as described above. This suggests that TGF β 2 signaling may be involved in dorsoventral regulation independent of *Shh* induction. Taken together, among the candidate genes we identified, WNT10B via the canonical WNT pathway and FGF2 via FGFR1 appear to regulate *Shh* induction and the subsequent patterning process.

WNT signaling may be the key for the dorsal properties in limb formation. It has been reported that canonical WNT/ β -catenin signaling plays essential roles in limb regeneration among vertebrates, including axolotls (Kawakami et al., 2006 [↗](#); Lovely et al., 2022 [↗](#)). In the present study, we demonstrated that introducing *Wnt10b* or BIO treatment could induce *Shh* expression in VentBL, facilitating limb patterning. In mouse limb development, *Wnt10b* is expressed in the apical ectodermal ridge (AER, Witte et al., 2009 [↗](#)), and mutations in *Wnt10b* are associated with Split-Hand/Foot Malformation (Ugur and Tolun, 2008 [↗](#); Al Ghamdi et al., 2020 [↗](#); Bilal et al., 2023 [↗](#)). However, there is no evidence suggesting that *Wnt10b* contributes to dorsal specificity during limb development. This raises questions about whether the function of *Wnt10b* is unique to axolotls or whether its orthologs in other species might share functional redundancy with other WNT genes. Further studies are needed to clarify this point. Notably, WNT10B utilizes the canonical Wnt signaling pathway, similar to WNT7A, a well-known WNT family gene critical for dorsoventral limb patterning in amniotes. In amniotes, *Wnt7a* is expressed in the dorsal ectoderm of developing limb buds and induces *Lmx1b* expression in dorsal mesenchyme, thereby establishing dorsal characteristics (Riddle et al., 1995 [↗](#); Cygan et al., 1997 [↗](#); Chen, 2002 [↗](#)). In axolotls, dorsal-specific *Wnt7a* expression has not been confirmed. Our RNA-seq analysis showed that there was almost no significant difference in *Wnt7a* expression levels between DorBL and VentBL ($\log_2$ (normalized counts) = 5.97, $\log_2$ (fold change) = -0.810, $p = 0.401$), consistent with previous studies (Shimokawa et al., 2013 [↗](#)). Similarly, there was no significant difference in *En1* expression ($\log_2$ (normalized counts) = 0.909, $\log_2$ (fold change) = -1.812, $p = 0.264$). In amniote limb development, *En1* is expressed in the ventral ectoderm, where it restricts *Wnt7a* expression to the dorsal ectoderm and thereby prevents induction of *Lmx1b* in the ventral mesenchyme (Loomis et al., 1996 [↗](#); Logan et al., 1997 [↗](#); Chen and Johnson, 2002 [↗](#)). These results suggest that *Wnt7a* does not have a dorsal-specific function, at least in axolotl limb regeneration. While WNT10B could function as a dorsal-mediated signal, WNT10B is unlikely to induce dorsal identity, as ectopic *Lmx1b* expression was not observed in *Wnt10b*-introduced VentBL, which formed double-ventral limbs (Fig. 5F–H [↗](#)). This indicates that *Wnt10b* in axolotl limb regeneration does not simply replace the function of *Wnt7a* in amniote limb development. Nevertheless, the involvement of canonical WNT signaling in dorsal function and limb morphogenesis remains an important area for further investigation.

We identified FGF2 as the ventral-mediated signal, which plays a crucial role in the limb patterning process during axolotl limb regeneration. This aligns with previous studies showing that *Fgf2* expression correlates with limb regeneration in salamanders (Giampaoli et al., 2003 [↗](#)). *Fgf2* expression was relatively high in VentBL, and *Fgf2*-introduced DorBL formed patterned limbs (Fig. 5 [↗](#)). These results indicate that *Fgf2* overexpression can substitute for the presence of ventral cells in the limb patterning process. It is noteworthy that FGF2 application does not appear to induce ventral identity, as the limbs formed from *Fgf2*-introduced DorBL exhibited dorsally symmetric limb structures. The use of FGF signaling as a ventral-mediated output downstream of

dorsoventral identity has not been documented in other species examined to date. Whether this represents an axolotl-specific regulatory mechanism or a broader function of FGF signaling in limb morphogenesis remains to be determined.

Our findings suggest that although WNT10B and FGF2 act as dorsal- and ventral-mediated signals, they do not alter dorsal or ventral identity itself. In amniote limb development, WNT7A and EN1 regulate dorsoventral identity through *Lmx1b* expression. In contrast, in axolotls, *Wnt10b*-electroporation in VentBL or *Fgf2*-electroporation in DorBL did not affect the expression patterns of *Lmx1b* (Fig. 5C). Moreover, the limbs formed from such DorBL and VentBL exhibited dorsally or ventrally symmetric structures (Fig. 5H–J, Fig. S4). These findings suggest that dorsoventral identities in axolotls are not affected by *Wnt10b* or *Fgf2* overexpression. We previously reported that the expression patterns of *Lmx1b* in axolotl limb regeneration are likely to depend on the positional origins of cells (Iwata et al., 2020; Yamamoto et al., 2022). The identities of cells along the dorsoventral axis may be controlled by their positional memory and determined before the initiation of limb regeneration.

The present study revealed that the dorsal- and ventral-mediated signals WNT10B and FGF2 regulate *Shh* expression during limb patterning in axolotls. These findings highlight a degree of conservation between axolotl limb regeneration and amniote limb development. In amniote limb development, FGF and WNT signaling pathways are key regulators of *Shh* expression. In developing amniote limb buds, *Fgf2* is expressed in the ectoderm, including the AER, and in adjacent mesoderm, and promotes distal outgrowth. FGFs, including FGF2, supplied from the AER are known to maintain *Shh* expression (Laufer et al., 1994; Niswander et al., 1994; Yang and Niswander, 1995; Li et al., 1996). Similarly, WNT7A regulates *Shh* expression in amniote limb development, and loss of WNT7A function results in reduced *Shh* expression in the zone of polarizing activity (ZPA) and the deletion of posterior structures (Parr and McMahon, 1995). Thus, our findings provide new insight into both conserved and divergent aspects of dorsal- and ventral-mediated signaling in the regulation of *Shh* expression, thereby furthering our understanding of limb morphogenesis.

In this study, RNA-seq analysis of ALM blastemas induced on the dorsal or ventral side listed *Wnt10b* and *Fgf2* as genes that are more highly expressed in DorBL and VentBL, respectively. In the ALM context, *Wnt10b* or *Fgf2* overexpression was sufficient to substitute for dorsal or ventral tissues, respectively, and to drive *Shh* induction and subsequent limb patterning even in the absence of those tissues (Fig. 5). However, in amputation-induced blastemas during normal regeneration, *in situ* hybridization did not reveal clear expression patterns for *Wnt10b* or *Fgf2* (Fig. S5A), and reanalysis of single-cell RNA-seq from the regenerating blastema (Li et al., 2021) showed that their expression did not strictly follow *Lmx1b* expression (Fig. S6J, K). By contrast, and consistent with our bulk RNA-seq results, RT-qPCR of manually microdissected dorsal and ventral halves of regenerating blastemas showed that *Wnt10b* and *Fgf2* were expressed at significantly higher levels in the dorsal and ventral halves, respectively (Fig. S5B). These results suggest that *Wnt10b* expression and *Fgf2* expression are mediated by dorsal and ventral cells, respectively, but their expression is not restricted to dorsal or ventral cells. Our results on *Wnt10b*-electroporated VentBL and *Fgf2*-electroporated DorBL suggest that both activation of WNT10B or FGF2 is required for *Shh* induction and proceed with limb patterning. To fully understand axolotl limb regeneration, it will be important to determine how *Wnt10b* and *Fgf2* expression is regulated, how their downstream programs are deployed, and how dorsal and ventral cells respond to these signals in future studies.

The dorsoventral-mediated *Shh* induction mechanism

In the present study, we found the importance of the co-presence of cells carrying the anterior, posterior, dorsal, and ventral identity for successful axolotl limb patterning (Fig. 7). In our proposed model, following amputation, nerves first trigger blastema induction by secreting nerve factors, such as BMPs and FGFs (Makanae et al., 2014; Satoh et al., 2016). These nerve factors stimulate connective tissue cells, including dermal cells, to generate multipotent mesenchymal blastemal cells (Muneoka et al., 1986; Kragl et al., 2009; Hirata et al., 2010; Gerber et al.,

2018). Within the induced blastema, *Wnt10b* and *Fgf2* expression are mediated by the dorsal and ventral cells, respectively. In the next phase, the co-existence of WNT10B and FGF2 signaling induces *Shh* expression in the posterior region of the blastema. During this phase, *Fgf8* is expressed in the anterior mesenchyme independently of the co-existence of dorsal and ventral cells (Fig. 7A). These expression patterns of *Wnt10b*, *Fgf2*, *Shh*, and *Fgf8* may be mediated by the positional memory of cells (Otsuki et al., 2021). In the subsequent phase, the anteroposterior interaction, mediated by FGF8 and SHH, supports distal outgrowth to form a limb. This model explains previous observations in studies on double-half limbs and ALM blastemas (AntBL, PostBL, DorBL, and VentBL), which typically fail to form a limb. In blastemas induced from double-anterior and double-posterior limbs, or AntBL (Fig. 7B) and PostBL (Fig. 7C), SHH or FGF8 proteins are likely absent due to the lack of posteriorly or anteriorly derived cells, respectively. Similarly, in blastemas induced by amputating double-dorsal and double-ventral limbs, or DorBL (Fig. 7D) and VentBL (Fig. 7E), FGF2 or WNT10B proteins are likely absent or insufficient because of the lack of ventrally or dorsally derived cells, respectively. This results in the absence of SHH, even if posteriorly derived cells are present. In all these cases, the absence of either FGF8 or SHH disrupts the anteroposterior interaction, leading to failure in limb patterning. We conclude that the requirement for cells derived from all four positional origins is underpinned by this model. In this model, dorsal- and ventral-mediated signals activate the posterior SHH, enabling mutual interaction with the anterior FGF8. This interplay ensures proper anteroposterior interaction and complete limb patterning. Our findings contribute to understanding how the integration of four positional identities—dorsal, ventral, anterior, and posterior—drives proper limb patterning during axolotl limb regeneration.

Materials and methods

Animal procedures

Axolotls between 4 and 15 cm from snout to tail tip were housed in tap water at 22°C. Both forelimbs and hind limbs were used for surgical procedures without distinction. Transgenic axolotls were obtained from the Ambystoma Genetic Stock Center (<http://www.ambystoma.org/genetic-stock-center>). Animals were anesthetized in 0.1% MS222 (Sigma-Aldrich) solution before surgical procedures.

In the ALM experiments, skin was peeled from the anterior, posterior, dorsal, or ventral side of a limb at the stylopod level, so that the skin of the opposite side was not injured. Thus, the size of the injured area depended on the limb size. Then, thick bundles of nerve trunks running the center of a limb was deviated the injured region. For the cell-tracing experiment, a piece of posterior half of the dorsal or ventral skin was obtained from GFP-expressing transgenic animals and grafted to the ALM region.

Protein-soaked beads for grafting were prepared as previously described (Makanae et al., 2014; Kashimoto et al., 2023). In brief, air-dried gelatin beads were allowed to swell in stock solution (1 µg/µL) prepared following the manufacturer's instructions. Equal amounts of proteins were used when formulating the combination protein mixture. Beads were soaked in the mixture of proteins (Bmp2 [mouse], Fgf2 [mouse], and Fgf8 [human/mouse]; R&D Systems) overnight at 4°C. For control experiments, gelatin beads were soaked in DDW. Before grafting, dorsal or ventral skin was peeled because full-thickness skin inhibits limb regeneration (Thornton, 1962; Mescher, 1976; Tsai, 2020). The beads were grafted under the wounded epidermis at 3 dps. The limbs were fixed 10 days after grafting. The details of grafting procedures are as previously described (Makanae et al., 2014; Kashimoto et al., 2023).

To analyze skeletal patterns, induced limbs were stained with alcian blue and alizarin red. Samples were fixed in 100% ethanol for 1 day at room temperature, then stained with Alcian blue solution (Wako, pH 2.0) in 20% acetic acid (Nacalai Tesque) with 80% ethanol solution at 37°C overnight. Then, samples were washed in tap water several times and fixed in 10% Formaldehyde Neutral Buffer Solution (Nacalai Tesque) for 1 day. Samples were then stained with Alizarin red S

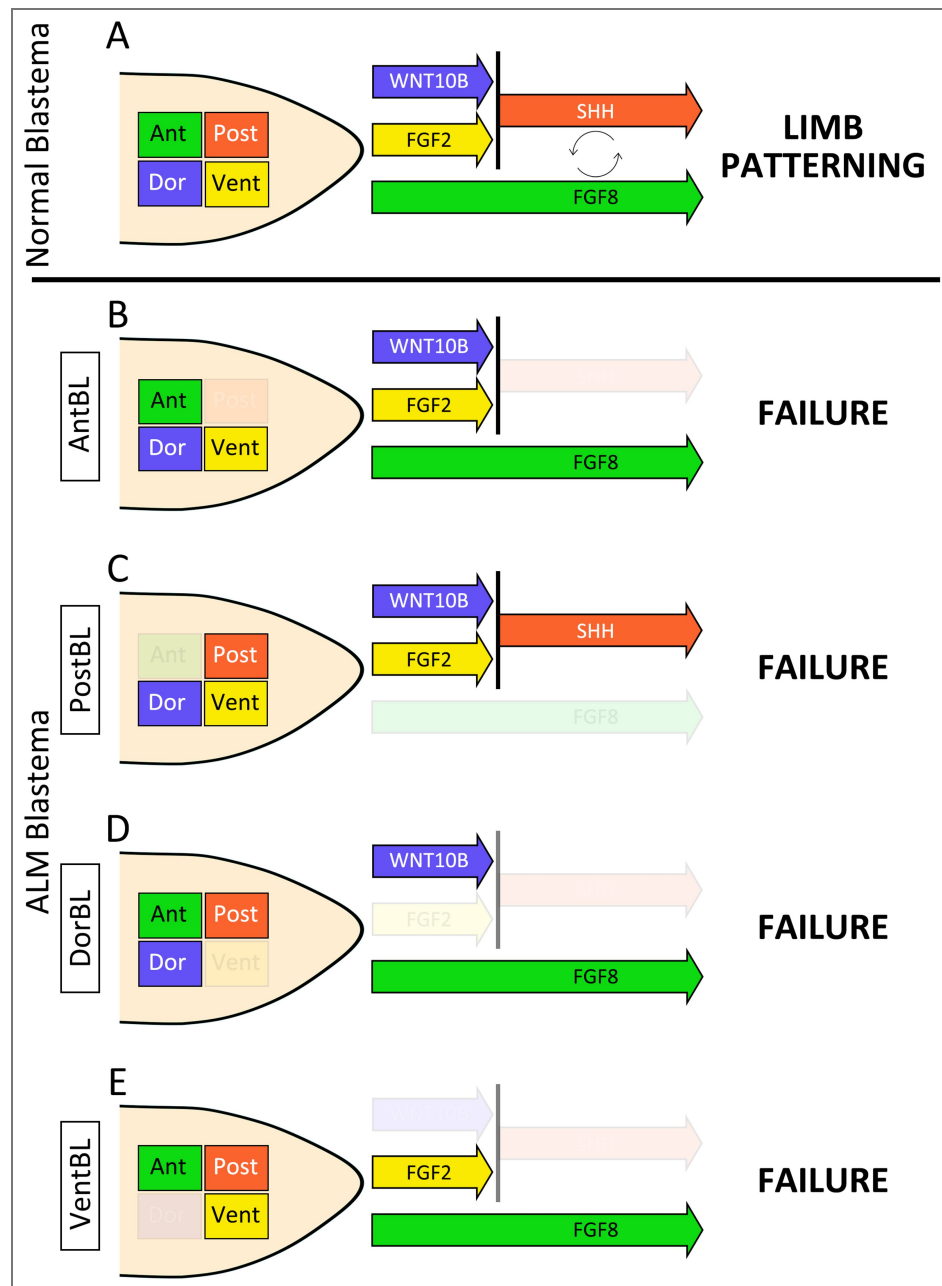


Figure 7. The dorsoventral-mediated Shh induction mechanism

Schematic images of a normal blastema (A), AntBL (B), PostBL (C), DorBL (D), and VentBL (E). Green, red, blue, and yellow boxes within the blastema represent cells derived from anterior, posterior, dorsal, and ventral regions, respectively. Colored arrows indicate the presence of the corresponding signals mediated by these cells. In this model, limb patterning requires both FGF8 and SHH, and *Shh* expression in posteriorly derived cells is induced by the co-existence of WNT10B and FGF2.

(Nacalai Tesque) in the solution (4% KOH:10% Formaldehyde Neutral Buffer Solution = 2:3) at room temperature for 1 day. Finally, samples were placed in graded glycerol with 4% KOH for clearing.

For BIO treatment, 10 mM stock solution of 6-bromoindirubin-3-oxime (BIO, Selleck, S7198) dissolved in DMSO was stored in the dark at 4°C. Axolotls soon after surgery were raised in tap water with BIO solution (experimental) or with the same amount of solvent, DMSO (control), until 14 dps for RT-qPCR and 30 dps for phenotype analysis. The containers including water and axolotls were kept in the dark during BIO or DMSO treatment. For BIO treatment, axolotls between 4 and 5 cm from snout to tail tip were used.

Sectioning and histological staining

Samples were fixed in 4% PFA/PBS overnight at room temperature before sectioning. The fixed samples were embedded in O.C.T. compound (Sakura) following 30% sucrose/PBS treatment for approximately 12 h at 4°C. Frozen sections of 14 µm thickness were prepared using Leica CM1850 (Nussloch). The sections were well dried under an air dryer and kept at -80°C until use.

Standard hematoxylin and eosin (HE) staining and Masson's trichrome staining were used for histological analysis. To visualize cartilage formation, Alcian blue staining was performed before HE staining. In brief, sections were washed in tap water several times to remove the O.C.T. compound. Then, the sections were stained with Alcian blue (Wako, pH 2.0), and then HE staining was performed. Trichrome stain (Masson) kit (Sigma, HT15-1KT) was used for trichrome staining. The stained sections were mounted using Softmount (Wako).

In situ hybridization

In situ hybridization was performed as described previously (Yamamoto et al., 2022 [DOI](#)). For probe synthesis, target genes cloned on pTAC-2 plasmid (BioDynamics Lab Inc.) were amplified by PCR using M13 primers. PCR fragments were purified and used as an RNA probe template. RNA probe synthesis was performed with Sp6 or T7 RNA polymerase (Takara) for 3 h, and RNA was hydrolyzed, depending on the length of targets. The sections were washed in PBT to remove the O.C.T. compound, treated with proteinase K (10 µg/mL) (Invitrogen)/PBT for 20 min at room temperature, washed in PBT, treated with 4% PFA/PBS for 20 min at room temperature, washed in PBT, and then probes were hybridized at 62.5°C for approximately 18 h. The sections were washed in wash buffer 1 (formamide:H₂O:20× SSC [3M NaCl:0.3M sodium citrate, pH 5.0] = 2:1:1), and then in wash buffer 2 (formamide:H₂O:20× SSC = 5:1:4). The samples were then incubated with anti-digoxigenin-AP Fab fragments (Sigma-Aldrich, 1/1000) for 2 h at room temperature. Samples were stained with BCIP (Nacalai Tesque) and NBT (Nacalai Tesque) in alkaline phosphatase buffer (0.1M NaCl, 0.1M Tris-HCl [pH9.5], 0.1% Tween20) for 18 h at room temperature after washing in TBST.

Immunofluorescence

Immunofluorescence on sections was carried out based on a previous report (Yamamoto et al., 2022 [DOI](#)). The antibodies were as follows: anti-GFP (MBL, #598, 1/500), anti-acetylated alpha tubulin (Santa Cruz, #sc-23950, 1/1000), anti-mouse IgG Alexa 488 (Invitrogen, #A11017, 1/1000), anti-rabbit IgG Alexa 488 (Invitrogen, #A21206, 1/500). Nuclei were stained with Hoechst (Nacalai Tesque), and images were captured using an Olympus BX51 system.

RNA-Seq analysis

Total RNA was extracted from DorBL and VentBL at 10 dps using TriPure reagent (Roche), following the manufacturer's instructions. Three biological replicates were prepared for both samples. 150 bp paired-end RNA-seq reads were obtained under contract to Rhelixa (Tokyo, Japan), using Illumina Nova Seq 6000 and SMART-Seq HT Plus Kit (#R400449). RNA-seq data were analyzed on Galaxy (<https://usegalaxy.eu/root> [DOI](#)) as follows. Sequence reads were trimmed and the quality was filtered by Trimmomatic v0.39 (Bolger et al., 2014 [DOI](#)) with the following parameters (LEADING:20, TRAILING:20, SLIDINGWINDOW:4:15, MINLEN:36). Axolotl genome data (AmexG_v6.0-DD) and annotation data (AmexT_v47-AmexG_v6.0-DD.gtf) were obtained from

AXOLOTL-OMICS (<https://www.axotl-omics.org/>) (Schloissnig et al., 2021). Mapping to the Axolotl genome (AmexG_v6.0-DD) was performed with HISAT2 v2.2.1 (Kim et al., 2015). Count data were obtained with featureCounts v2.0.8 (Liao et al., 2013) on the basis of the Axolotl gene model (AmexT_v47-AmexG_v6.0-DD.gtf). DEGs were identified with DESeq2 v2.11.40.8 (Love et al., 2014) ($p < 0.05$). Among identified DEGs (DorBL > VentBL; 762 genes, VentBL > DorBL; 513 genes), genes annotated as “intercellular signaling molecule” were explored on PANTHER (<https://www.pantherdb.org/>) (Thomas et al., 2003), and 21 genes (DorBL > VentBL; 5 genes, VentBL > DorBL; 16 genes) were identified as candidate genes of dorsal- and ventral-mediated signals, as shown below:

DorBL > VentBL:

WNT4 (AMEX60DD052091), WNT10B (AMEX60DD029981), SEMA3A (AMEX60DD023165), LECT2 (AMEX60DD041044), ANGPTL5 (AMEX60DD049512).

VentBL > DorBL:

FCN1 (AMEX60DD050926), CXCL14 (AMEX60DD028973), DNER (AMEX60DD002010), FGF7 (AMEX60DD003767), CXCL12 (AMEX60DD052412), CXCL8 (AMEX60DD044133), SEMA6A (AMEX60DD042813), ANGPTL2 (AMEX60DD050490), ANGPTL6 (AMEX60DD031854), FGL2 (AMEX60DD006387), VWDE (AMEX60DD022491), TGFB2 (AMEX60DD036126), FGF2 (AMEX60DD044865), EFNA4 (AMEX60DD014882), EFNA3 (AMEX60DD014867), ANGPTL1 (AMEX60DD018261).

Cloning candidate genes and preparation of construct vectors for overexpression

Among the identified candidate genes, we focused on *Wnt10b*, *Wnt4*, *Fgf2*, *Fgf7*, and *Tgfb2*. We cloned these genes in pTAC-2 plasmids from cDNA obtained from blastemas. The following primers were used:

Wnt10b Fwd:ATGGCCACAGCTCACCCTCCGACACC
Wnt10b Rev:TCACTTGACACATTACCCATTCTGTG
Wnt4 Fwd:ATGGATGCTCACGAAAGCAGCGTATATC
Wnt4 Rev:TCACCGGCAGGTGTGCATTCTACCAC
Fgf2 Fwd:ATGGCGGCGGGGAGCATCACACCTTGC
Fgf2 Rev:TCAACTCTTGCCGACATGGGAAGGAAAAAG
Fgf7 Fwd:ATGCGCAGATGGGTGCTAGCTTGGATC
Fgf7 Rev:TCATGTGTTATTGGATATACGCATTGGA
Tgfb2 Fwd:ATGAGATTACAATTACTGAGAAAAAAAATG
Tgfb2 Rev:TTAGCTGCACTTGCAAGATTTTACAATCA

We then inserted these genes to p2a-AcGFP-pCS2 vectors with In-Fusion® HD Cloning Kit (Clontech). The following primers were used for PCR before In-Fusion reaction:

pCS2 inverse Fwd:GCTACTAACTTCAGCCTGCTGAAGCAGG
pCS2 inverse Rev:CATCGATGGGATCCTGCAAAAAGAACAGTAGCTT
Wnt10b Fwd:AGGATCCCATCGATGGCCACAGCTCACCCTCCGA
Wnt10b Rev:GCTGAAGTTAGTAGCCTTGACACATTACCCA
Wnt4 Fwd:AGGATCCCATCGATGGATGCTCACGAAAGCAGCG
Wnt4 Rev:GCTGAAGTTAGTAGCCCGGAGGTGTGCATTCTA
Fgf2 Fwd:AGGATCCCATCGATGGCGGCGGGGAGCATCACC
Fgf2 Rev:GCTGAAGTTAGTAGCACTCTTGCCGACATGGGAA
Fgf7 Fwd:AGGATCCCATCGATGCGCAGATGGGTGCTAGCTTG
Fgf7 Rev:GCTGAAGTTAGTAGCTGTGTTATTGGATATACGA
Tgfb2 Fwd:AGGATCCCATCGATGAGATTACAATTACTGAGAAA
Tgfb2 Rev:GCTGAAGTTAGTAGCGCTGCACTTGCAAGATTTA

Finally, the following DNA constructs were obtained; *Wnt10b*-p2a-GFP (pCS2), *Wnt4*-p2a-GFP (pCS2), *Fgf2*-p2a-GFP (pCS2), *Fgf7*-p2a-GFP (pCS2), and *Tgfb2*-p2a-GFP (pCS2).

Electroporation

Each DNA construct was injected directly into the target region. Immediately after injection, electric pulses were applied (20 V, 50 ms pulse length, 950 ms interval, 10 times) with NEPA21(Nepa gene). The injected DNA constructs were as follows: pCS2-AcGFP, pCS2-*Shh*-p2a-AcGFP, *Wnt10b*-p2a-GFP (pCS2), *Wnt4*-p2a-GFP (pCS2), *Fgf2*-p2a-GFP (pCS2), *Fgf7*-p2a-GFP (pCS2), and *Tgfb2*-p2a-GFP (pCS2). All plasmids were purified using a Genopure Maxi kit (Roche). Electroporation was performed at -3, 4, and 7 dps and samples were fixed at 10 dps for *in situ* hybridization. To analyze the phenotype of 90 dps samples, electroporation was performed at -3, 4, 7, 10, 15, 20, 25, and 30 dps.

qRT-PCR

The procedures of qRT-PCR were previously described (Yamamoto et al., 2022 [\[1\]](#)). In brief, RT was performed using Prime Script II (Takara), and RT-qPCR was performed using KAPA SYBR FAST qPCR Master Mix (Kapa Biosystems) and StepOne (ThermoFisher Scientific). Primers were as follows:

Ef-1α Fwd: AACATCGTGGTCATCGGCCAT Ef-1α
 Rev: GGAGGTGCCAGTGATCATGTT
 Shh Fwd: GCTCTGTGAAAGCAGAGAACTCG
 Shh Rev: CGCTCCGTCTCTATCACGTAGAA
 Axin2 Fwd: GGCATGACTTATCCCCAGG
 Axin2 Rev: GCATCATTGGCTGTCAACGG
 Lef1 Fwd: CTACACCGAGATCAGCCACC
 Lef1 Rev: GCTGTGGTAGGAGTTGTGGG
 Lmx1b Fwd: CTGGTCCATGGCTACGATCT
 Lmx1b Rev: TTAGCAGCAGAAACGGGACT
 Wnt10b Fwd: CAGAAGAGACCCAGGTGCAG
 Wnt10b Rev: CGAAGGCCCAAGATGTCTGT
 Fgf2 Fwd: TCTTCCTTCGCATCAACCCC
 Fgf2 Rev: TTTCATTGCCATCAACCGCC

RNA for Fig. S3E [\[1\]](#) was prepared from 1μM BIO- or DMSO-treated VentBL at 14 dps. *Ef-1α* was used as the internal control.

Pixel classification and calculating symmetry scores

A machine learning-based method was applied for pixel classification using ilastik software (downloaded on <https://www.ilastik.org/> [\[2\]](#)). The details and workflows were previously described (Berg et al., 2019 [\[3\]](#)). Each pixel in the images was classified into five classes. The regions of the background (Class 1), cartilage (Class 2), muscle (Class 3), other connective tissue (Class 4), and epidermis (Class 5) were annotated for training data. Using these annotated regions as references, pixels were automatically classified into the respective classes. In this process, the same training data were used for images obtained from the same limb. Then, symmetry scores were calculated for each class individually and for the combined set of all classes. In this process, the external shape of the section, which could bend randomly during sample fixation process, would affect the symmetry scores if the entire region were used. To focus on the symmetry of the anatomical patterns, images with 400 μm width were prepared (Fig. 4L [\[4\]](#), 5E, F [\[5\]](#)). The symmetry scores of pixel-classified images were calculated using python. First, the center of the dorsal end and ventral end was set as the axis of symmetry. Then, one side of the image was flipped. Next, color masks were generated for both sides of the image. These masks identified pixels that matched the specified color. Pixels were considered to match if their RGB values were within the given tolerance range for all three channels. In pixel comparison, the masks for one side and the other side were compared to calculate the following pixels:

Matching pixels: The number of pixels that match in both sides for the specified color. Total pixels: The total number of pixels matching the color in either side.

The symmetry scores were computed as (Number of Matching Pixels)/(Total Pixels in Both Sides). The scores were obtained from 12 areas of each group. For statistical analysis, a two-tailed Welch's *t*-test was used. In this analysis, each group was compared to the intact limb group because the intact limb should be set as a typical asymmetrical structure.

Reanalyzing single-cell data

We reanalyzed published single-cell RNA-seq data from a 7 dpa (MB) blastema (Li et al., 2021 [DOI](#), Fig. S6 [DOI](#)). We constructed a transcriptome index for axolotl from the available genome assembly and gene annotation (AmexG_v6.0-DD genome and AmexT_v47-AmexG_v6.0-DD annotation, Schloissnig et al., 2021 [DOI](#)). The transcript set was used to build a salmon index with default k-mer length (31), and without decoy sequences, and gene-level UMI counts were generated using the Alevin module of salmon, which performs lightweight (quasi-)mapping of reads directly to the transcriptome index (Patro et al., 2017 [DOI](#); Srivastava et al., 2019 [DOI](#)). The mapping was done on Galaxy (<https://usegalaxy.eu> [DOI](#)). This procedure yielded a whitelist of 8,925 barcodes, corresponding to putative cells, and produced a per-cell by per-gene UMI count matrix. According to the Alevin summary output, 77.4% of reads aligned to the indexed axolotl transcriptome after barcode correction and UMI deduplication. The resulting count matrix was exported in Matrix Exchange (MEX) format (matrix.mtx, barcodes.tsv, features.tsv).

The data were then imported into R (RStudio environment) as a Seurat object, a data structure for scRNA seq data (Hao et al., 2021 [DOI](#)). Cells were filtered based on standard quality-control metrics, excluding droplets with extremely low gene complexity, extremely high total UMI counts, or unusually high mitochondrial or ribosomal RNA content. For normalization and integration of cells into a shared space, counts were log-normalized using Seurat's NormalizeData (default LogNormalize method with a scale factor of 10,000), and highly variable genes (HVGs) were identified using FindVariableFeatures (vst method; 2,000 features). The data were scaled with ScaleData, and principal component analysis (PCA) was performed on the HVGs (RunPCA, 50 components). The first 30 principal components were used to construct a nearest-neighbor graph (FindNeighbors) and to perform community detection-based clustering (FindClusters, resolution = 0.3). The same set of PCs (dims 1:30) was also used for nonlinear dimensionality reduction by UMAP (RunUMAP). Gene expression patterns were visualized using FeaturePlot with order = TRUE so that high-expressing cells are drawn on top of low- or non-expressing cells. We also generated two-gene "co-expression" maps by classifying cells as expressing gene A only, gene B only, both, or neither, and overlaying these classes on the UMAP using DimPlot.

Data availability

RNA-seq FASTQ files have been deposited in the DNA Data Bank of Japan (DDBJ; <https://www.ddbj.nig.ac.jp/> [DOI](#)) under BioProject accession PRJDB38065.

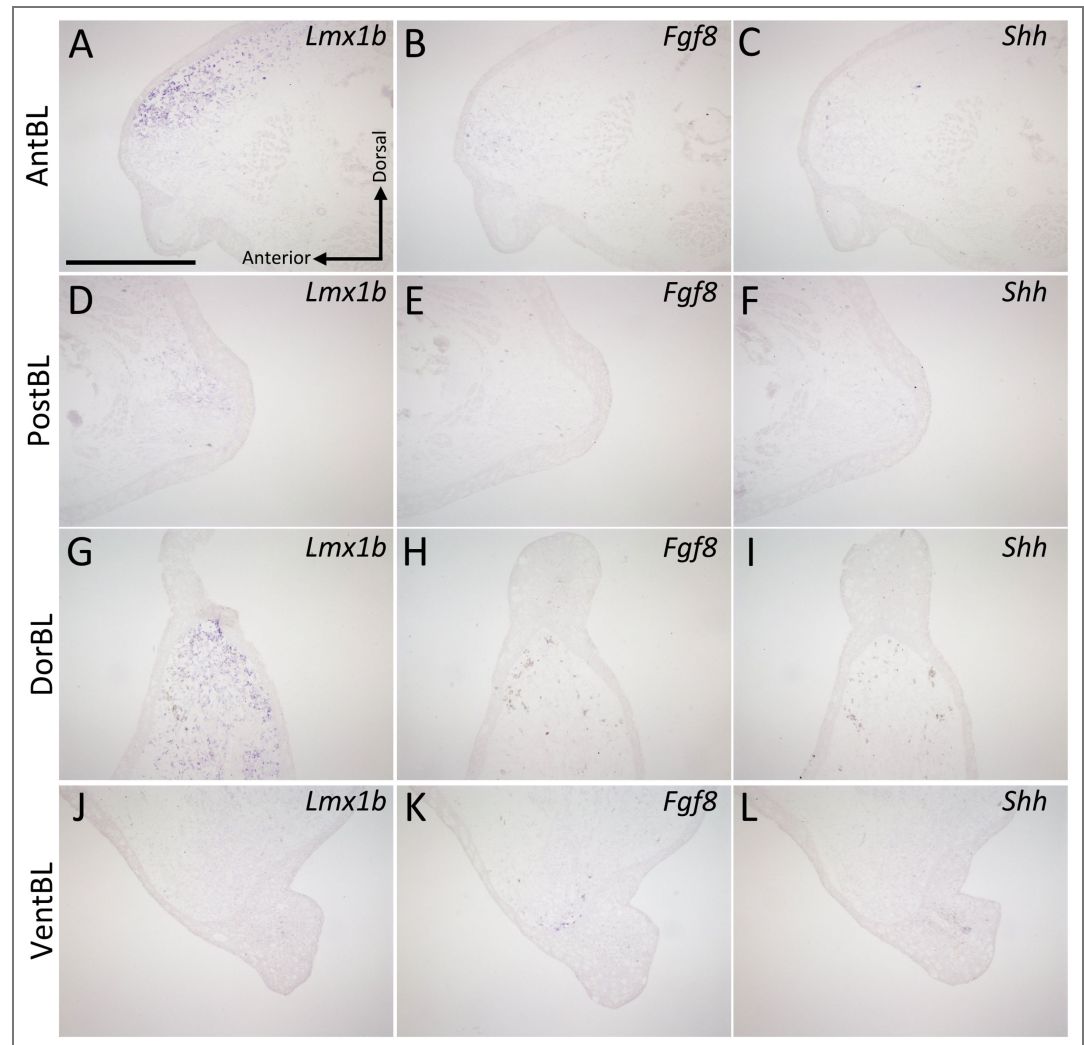


Figure S1. Gene expression patterns across the entire ALM-induced blastemas Sections of ALM blastemas induced at the anterior (A–C), posterior (D–F), dorsal (G–I), or ventral (J–L) region at 10 dps. *Lmx1b* (A, D, G, J), *Fgf8* (B, E, H, K), and *Shh* (C, F, I, L) expression was detected by *in situ* hybridization. These panels show the full blastema regions corresponding to the higher-magnification views in Fig. 2. Scale bar = 700 μ m (A).

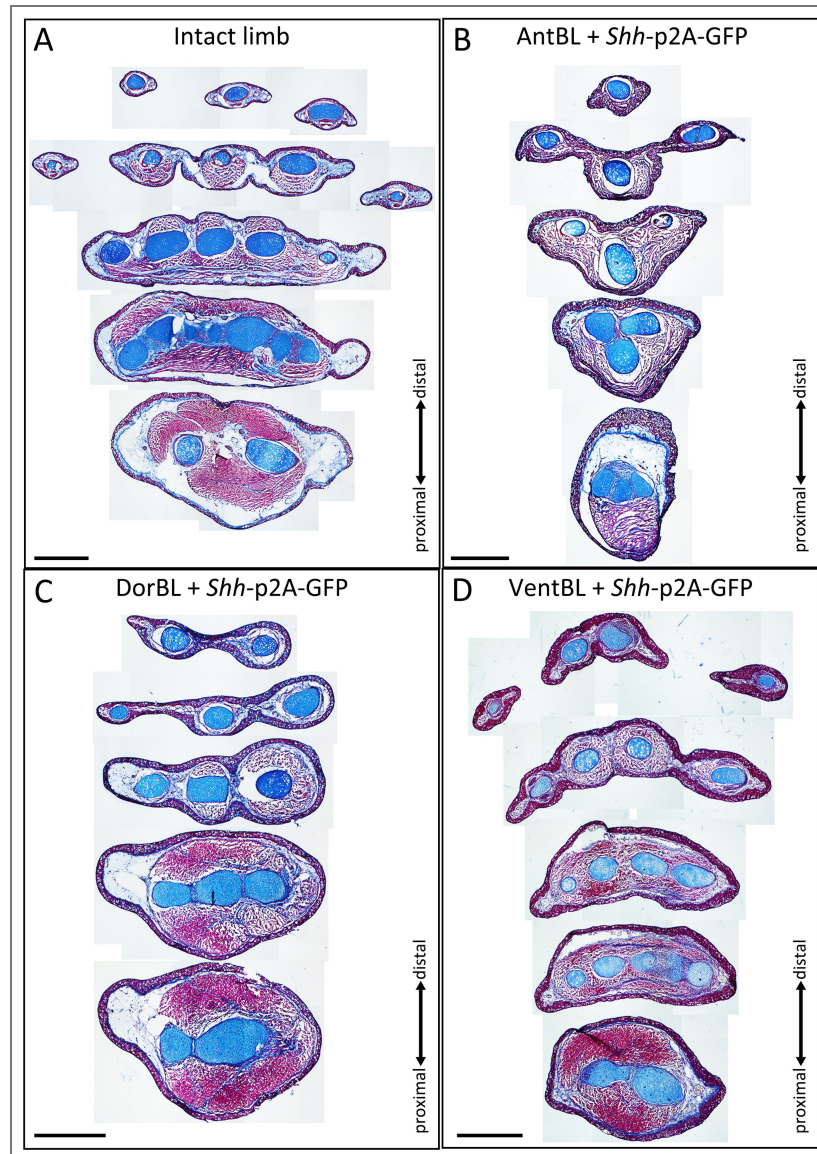


Figure S2. Histological analysis of intact limbs and limbs induced by Shh electroporation

(A–D) Transverse sections taken at multiple levels along the proximodistal axis from an intact limb and from the limbs shown in Fig. 4. Sections from limbs induced from DorBL and VentBL exhibit dorsoventrally symmetric internal structures, whereas intact limbs and limbs induced from AntBL do not. Scale bar = 1 mm.

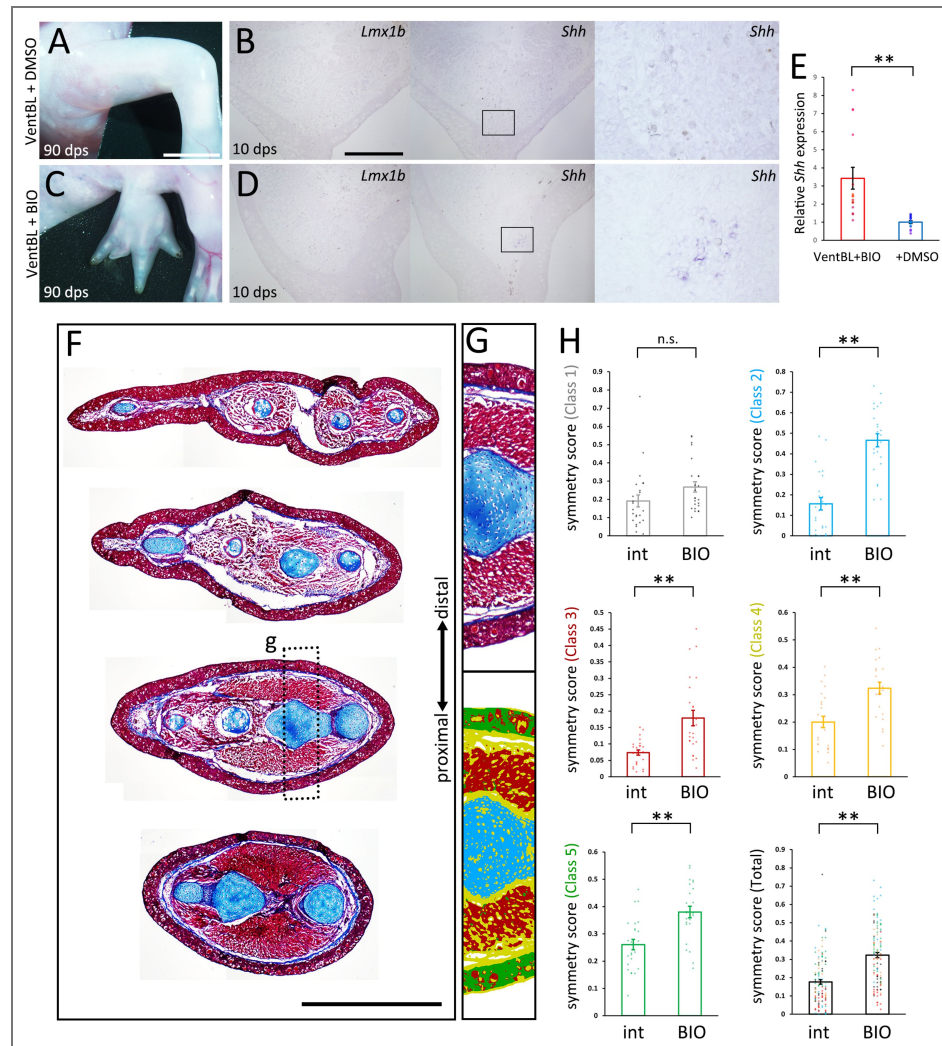


Figure S3. Limb patterning from BIO-treated VentBL

(A, C) VentBL treated with 1 μ M BIO (C) or DMSO (A) at 90 dps. (E) Quantitative analysis of *Shh* expression in VentBL treated with 1 μ M BIO and with DMSO. RNA was prepared from 4 identical VentBL for both. Technological replicates are plotted in the same color. (C–D) Histological analysis and pixel classification was performed in the same way as in Fig. 4. The dotted boxes indicate the regions shown in (D). (E) Symmetry scores of each class. Scores obtained from the same limb are plotted at the same x-coordinates. The plots of “int” are the same plots as in Fig. 4. n.s.; no significant difference, * p <0.05, ** p <0.005. Scale bar = 4 mm (B), 1 cm (C).

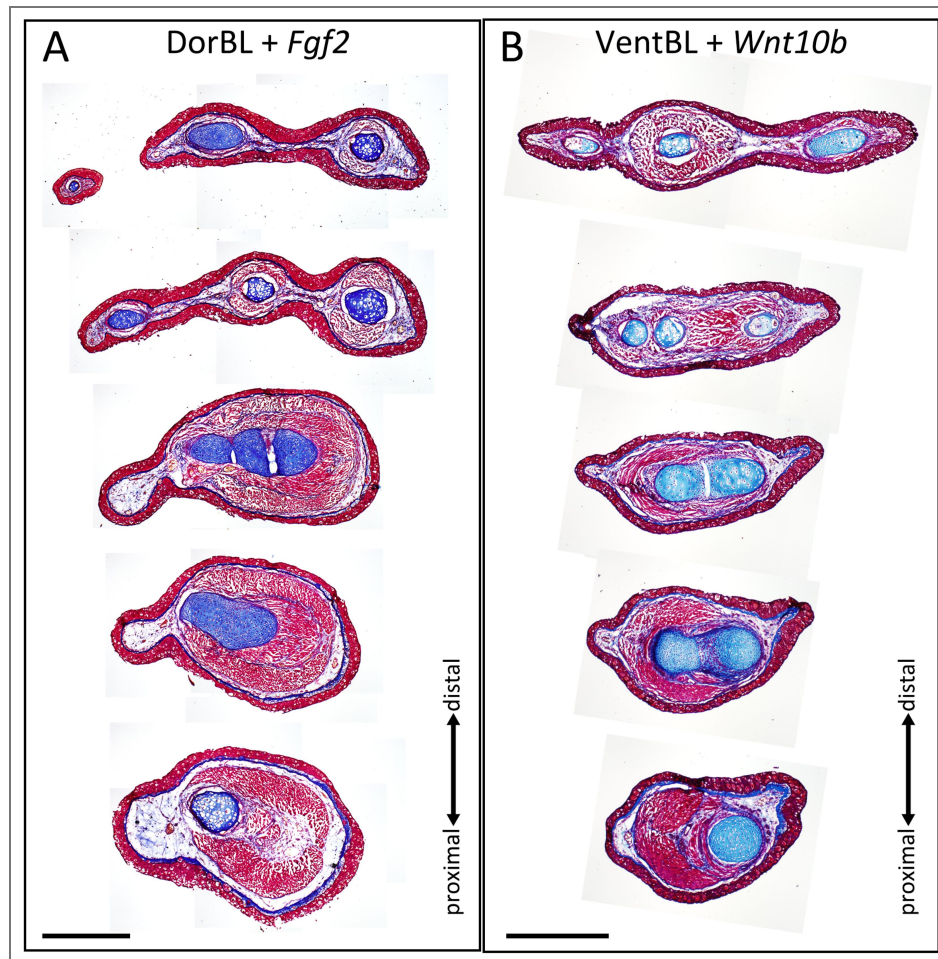


Figure S4. Histological analysis of limbs induced by *Fgf2* or *Wnt10b* electroporation

(A, B) Transverse sections taken at multiple levels along the proximodistal axis from the limbs shown in Fig. 5. The induced limbs exhibit dorsoventrally symmetric internal structures. Scale bar = 1 mm.

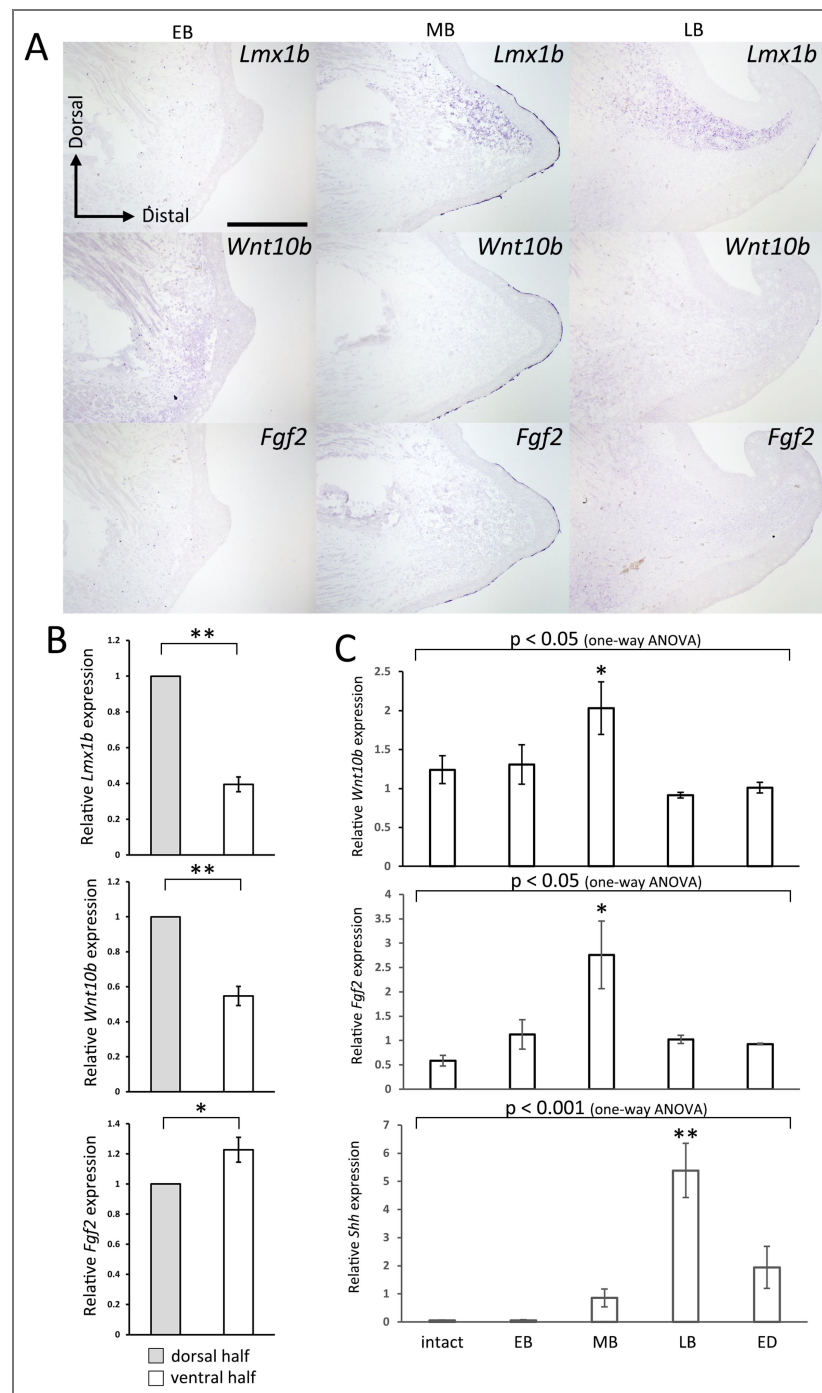


Figure S5. Gene expression patterns during normal limb regeneration

(A) Sections of amputation-induced blastemas at the early bud (EB), middle bud (MB), and late bud (LB) stages. *Lmx1b*, *Wnt10b*, and *Fgf2* expression was visualized by *in situ* hybridization. (B) Quantitative analysis of *Lmx1b*, *Wnt10b*, and *Fgf2* expression in MB-stage blastemas during normal regeneration (n=13). Gene expression was quantified by RT-qPCR on manually microdissected dorsal and ventral halves of each blastema. * $p < 0.05$, ** $p < 0.005$ (two-tailed paired *t*-test). (C) Quantitative analysis of *Wnt10b*, *Fgf2*, and *Shh* expression across stages (intact, EB, MB, LB, and early digit [ED]; n = 5 for all stages). *Wnt10b* and *Fgf2* expression peaked at the MB stage, whereas *Shh* expression peaked later, at the LB stage. One-way ANOVA detected a stage-dependent difference ($p < 0.05$ for *Wnt10b* and *Fgf2*; $p < 0.001$ for *Shh*), followed by Dunnett's test. * $p < 0.05$, ** $p < 0.005$.

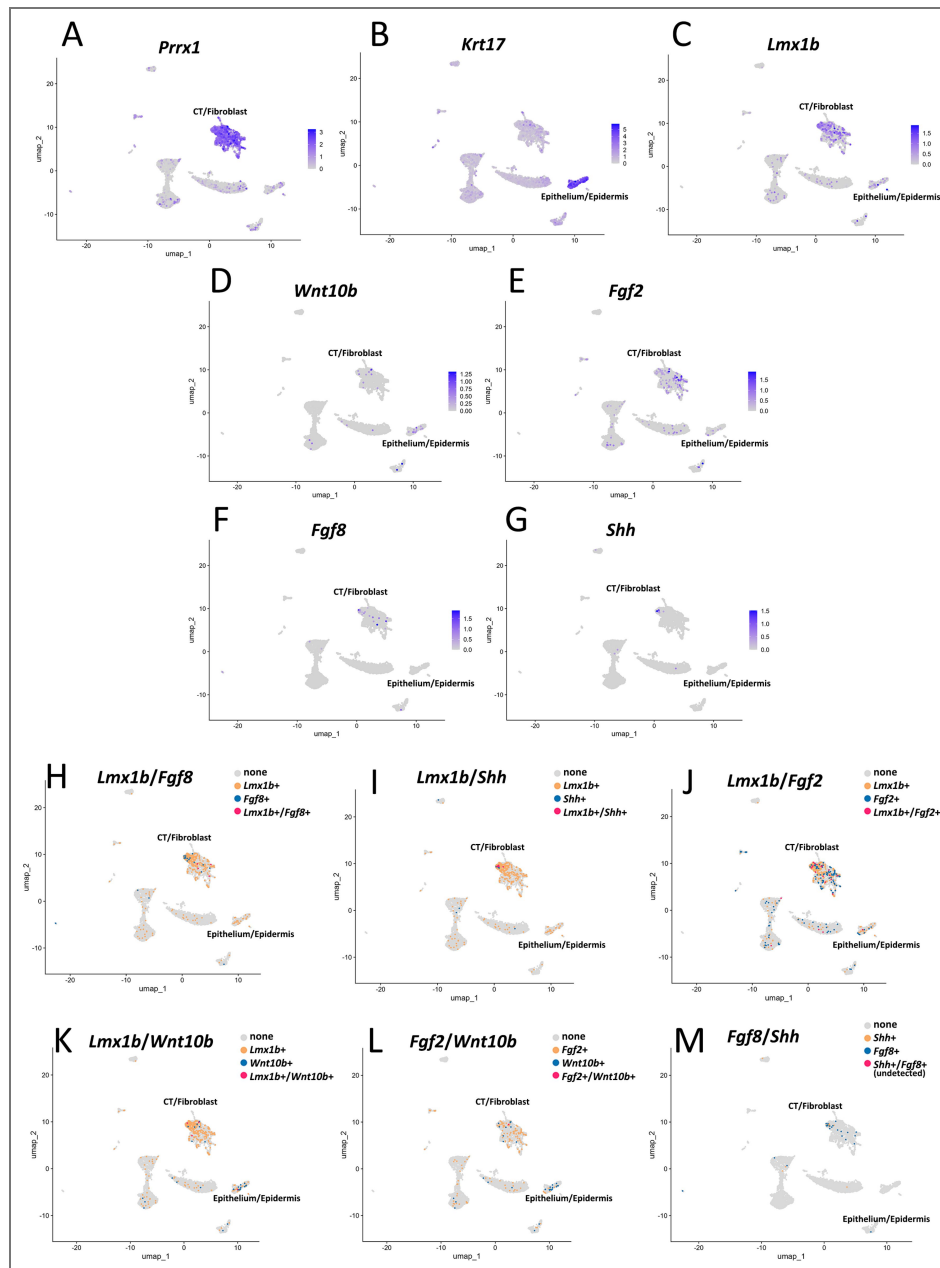


Figure S6. Gene expression patterns in a normal blastema assessed by reanalysis of axolotl single-cell RNA-seq data

Reanalysis of single-cell RNA-seq data from a middle bud (MB) stage blastema (Li et al., 2021). (A–G) FeaturePlot visualizations of *Prrx1* (mesenchyme marker), *Krt17* (epithelium marker), *Lmx1b*, *Wnt10b*, *Fgf2*, *Fgf8*, and *Shh*. *Lmx1b*, *Fgf2*, *Fgf8*, and *Shh* expression are detected in the mesenchymal cluster, whereas *Wnt10b* expression is detected in both mesenchymal and epithelial clusters. (H–M) “Co-expression” maps generated by classifying cells as expressing gene A only, gene B only, both genes, or neither gene, and overlaying these classes on the UMAP using DimPlot.

Acknowledgements

We are grateful to R. Iwata and T. Satoh for supporting office work and animal housing. Animals were obtained through Hiroshima University Amphibian Research Center. This work is supported by the Japan Society for the Promotion of Science KAKENHI grant-in-aid for scientific research (B) (20H03264 and 24K02034 to A.S.) and by a Grant-in-Aid for Japan Society for the Promotion of Science fellows (24KJ1718 to S.Y.).

Additional information

Author contributions

Conceptualization: S.Y., A.S.; Methodology: S.Y., M.H., A.S.; Validation: S.Y.; Formal analysis: S.Y.; Investigation: S.Y.; Resources: S.Y., S.F., A.O., A.S.; Data curation: S.Y.; Writing - original draft: S.Y.; Writing - review & editing: S.Y., S.F., A.O., M.H., A.S.; Visualization: S.Y., A.S.; Supervision: A.S.; Project administration: A.S.; Funding acquisition: A.S., S.Y.

Funding

Funder	Grant reference number	Author
Japan Society for the Promotion of Science	20H03264	Akira Satoh
MEXT Japan Society for the Promotion of Science (JSPS)	24K02034	Akira Satoh
MEXT Japan Society for the Promotion of Science (JSPS)	24KJ1718	Sakiya Yamamoto

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Peer reviews

Reviewer #1 (Public review):

Summary:

The manuscript by Yamamoto et al. presents a model by which the four main axes of the limb are required for limb regeneration to occur in the axolotl. A longstanding question in regeneration biology is how existing positional information is used to regenerate the correct missing elements. The limb provides an accessible experimental system by which to study the involvement of the anteroposterior, dorsoventral, and proximodistal axes in the regenerating limb. Extensive experimentation has been performed in this area using grafting experiments. Yamamoto et al. use the accessory limb model and some molecular tools to address this question. There are some interesting observations in the study. In particular, one strength the potent induction of accessory limbs in the dorsal axis with BMP2+Fgf2+Fgf8 is very interesting. Although interesting, the study makes bold claims about determining the molecular basis of DV positional cues, but the experimental evidence is not definitive and does not take into account the previous work on DV patterning in the amniote limb. Also, testing the hypothesis on blastemas after limb amputation would be needed to support the strong claims in the study.

Strengths:

The manuscript presents some novel new phenotypes generated in axolotl limbs due to Wnt signaling. This is generally the first example in which Wnt signaling has provided a gain of function in the axolotl limb model. They also present a potent way of inducing limb patterning in the dorsal axis by the addition of just beads loaded with Bmp2+Fgf8+Fgf2.

Comments on revised version:

Re-evaluation: The authors have significantly improved the manuscript and their conclusions reflect the current state of knowledge in DV patterning of tetrapod limbs. My only point of consideration is their claim of mesenchymal and epithelial expression of Wnt10b and the finding that Fgf2 and Wnt10b are lowly expressed. It is based upon the failed ISH, but this doesn't mean they aren't expressed. In interpreting the Li et al. scRNAseq dataset, conclusions depend heavily on how one analyzes and interprets it. The 7DPA sample shows a very low representation of epithelial cells compared to other time points, but this is likely a technical issue. Even the epithelial marker, Krt17, and the CT/fibroblast marker show some expression elsewhere. If other time points are included in the analysis, Wnt10b, would be interpreted as relatively highly expressed almost exclusively in the epithelium. By selecting the 7dpa timepoint, which may or may not represent the MB stage as it wasn't shown in the paper, the conclusions may be based upon incomplete data. I don't expect the authors to do more work, but it is worth mentioning this possibility. The authors have considered and made efforts to resolve previous concerns.

<https://doi.org/10.7554/eLife.106917.2.sa3>

Reviewer #2 (Public review):

Summary:

This study explores how signals from all sides of a developing limb, front/back and top/bottom, work together to guide the regrowth of a fully patterned limb in axolotls, a type of salamander known for its impressive ability to regenerate limbs. Using a model called the Accessory Limb Model (ALM), the researchers created early staged limb regenerates (called blastemas) with cells from different sides of the limb. They discovered that successful limb regrowth only happens when the blastema contains cells from both the top (dorsal) and bottom (ventral) of the limb. They also found that a key gene involved in front/back limb patterning, called Shh (Sonic hedgehog), is only turned on when cells from both the dorsal and ventral sides come into contact. The study identified two important molecules, Wnt10B and FGF2, that help activate Shh when dorsal and ventral cells interact. Finally, the authors

propose a new model that explains how cells from all four sides of a limb, dorsal, ventral, anterior (front), and posterior (back), contribute at both the cellular and molecular level to rebuilding a properly structured limb during regeneration

Strengths:

The techniques used in this study, like delicate surgeries, tissue grafting, and implanting tiny beads soaked with growth factors, are extremely difficult, and only a few research groups in the world can do them successfully. These methods are essential for answering important questions about how animals like axolotls regenerate limbs with the correct structure and orientation. To understand how cells from different sides of the limb communicate during regeneration, the researchers used a technique called *in situ* hybridization, which lets them see where specific genes are active in the developing limb. They clearly showed that the gene *Shh*, which helps pattern the front and back of the limb, only turns on when cells from both the top (dorsal) and bottom (ventral) sides are present and interacting. The team also took a broad, unbiased approach to figure out which signaling molecules are unique to dorsal and ventral limb cells. They tested these molecules individually and discovered which could substitute for actual dorsal and ventral cells, providing the same necessary signals for proper limb development. Overall, this study makes a major contribution to our understanding of how complex signals guide limb regeneration, showing how different regions of the limb work together at both the cellular and molecular levels to rebuild a fully patterned structure.

Weaknesses:

Because the expressional analyses are performed on thin sections of regenerating tissue, in the original manuscript, they provided only a limited view of the gene expression patterns in their experiments, opening the possibility that they could be missing some expression in other regions of the blastema. Additionally, the quantification method of the expressional phenotypes in most of the experiments did not appear to be based on a rigorous methodology. The authors' inclusion of an alternate expression analysis, qRT-PCR, on the entire blastema helped validate that the authors are not missing something in the revised manuscript.

Overall, the number of replicates per sample group in the original manuscript was quite low (sometimes as low as 3), which was especially risky with challenging techniques like the ones the authors employ. The authors have improved the rigor of the experiment in the revised manuscript by increasing the number of replicates. The authors have not performed a power analysis to calculate the number of animals used in each experiment that is sufficient to identify possible statistical differences between groups. However, the authors have indicated that there was not sufficient preliminary data to appropriately make these quantifications.

Likewise, in the original manuscript, the authors used an AI-generated algorithm to quantify symmetry on the dorsal/ventral axis, and my concern was that this approach doesn't appear to account for possible biases due to tissue sectioning angles. They also seem to arbitrarily pick locations in each sample group to compare symmetry measurements. There are other methods, which include using specific muscle groups and nerve bundles as dorsal/ventral landmarks, that would more clearly show differences in symmetry. The authors have now sufficiently addressed this concern by including transverse sections of the limbs and have explained the limitations of using a landmark-based approach in their quantification strategy.

<https://doi.org/10.7554/eLife.106917.2.sa2>

Reviewer #3 (Public review):

Summary:

After salamander limb amputation, the cross-section of the stump has two major axes: anterior-posterior and dorsal-ventral. Cells from all axial positions (anterior, posterior, dorsal, ventral) are necessary for regeneration, yet the molecular basis for this requirement has remained unknown. To address this gap, Yamamoto et al. took advantage of the ALM assay, in which defined positional identities can be combined on demand and their effects assessed through the outgrowth of an ectopic limb. They propose a compelling model in which dorsal and ventral cells communicate by secreting Wnt10b and Fgf2 ligands respectively, with this interaction inducing Shh expression in posterior cells. Shh was previously shown to induce limb outgrowth in collaboration with anterior Fgf8 (PMID: 27120163). Thus, this study completes a concept in which four secreted signals from four axial positions interact for limb patterning. Notably, this work firmly places dorsal-ventral interactions upstream of anterior-posterior, which is striking for a field that has been focussed on anterior-posterior communication. The ligands identified (Wnt10b, Fgf2) are different to those implicated in dorsal-ventral patterning in the non-regenerative mouse and chick models. The strength of this study is in the context of ALM/ectopic limb engineering. Although the authors attempt to assay the expression of Wnt10b and Fgf2 during limb regeneration after amputation, they were unable to pinpoint the precise expression domains of these genes beyond 'dorsal' and 'ventral' blastema. Given that experimental perturbations were not performed in regenerating limbs - almost exclusively under ALM conditions - this author finds the title "Dorsoventral-mediated Shh induction is required for axolotl limb regeneration" a little misleading.

Strengths:

- (1) The ALM and use of GFP grafts for lineage tracing (Figures 1-3) take full advantage of the salamander model's unique ability to outgrow patterned limbs under defined conditions. As far as I am aware, the ALM has not been combined with precise grafts that assay 2 axial positions at once, as performed in Figure 3. The number of ALMs performed in this study deserves special mention, considering the challenging surgery involved.
- (2) The authors identify that posterior Shh is not expressed unless both dorsal and ventral cells are present. This echoes previous work in mouse limb development models (AER/ectoderm-mesoderm interaction) but this link between axes was not known in salamanders. The authors elegantly reconstitute dorsal-ventral communication by grafting, finding that this is sufficient to trigger Shh expression (Figure 3 - although see also section on Weaknesses).
- (3) Impressively, the authors discovered two molecules sufficient to substitute dorsal or ventral cells through electroporation into dorsal- or ventral- depleted ALMs (Figure 5). These molecules did not change the positional identity of target cells. The same group previously identified the ventral factor (Fgf2) to be a nerve-derived factor essential for regeneration. In Figure 6, the authors demonstrate that nerve-derived factors, including Fgf2, are alone sufficient to grow out ectopic limbs from a dorsal wound. Limb induction with a 3-factor cocktail without supplementing with other cells is conceptually important for regenerative engineering.
- (4) The writing style and presentation of results is very clear.

Overall appraisal:

This is a logical and well-executed study that creatively uses the axolotl model to advance an important framework for understanding limb patterning. The relevance of the mechanisms to normal limb regeneration are not yet substantiated, in the opinion of this reviewer. Additionally, Wnt10b and Fgf2 should be considered molecules sufficient to substitute dorsal and ventral identity (solely in terms of inducing Shh expression). It is not yet clear whether these molecules are truly necessary (loss of function would address this).

Comments on revisions:

Congratulations - I still find this an elegant and easy-to-read study with significant implications for the field! Linking your mechanisms to normal limb regeneration (i.e. regenerating blastema, not ALM), as well as characterising the cell populations involved, will be interesting directions for the future.

<https://doi.org/10.7554/eLife.106917.2.sa1>

Author response:

The following is the authors' response to the current reviews.

Public Reviews:

Reviewer #1 (Public review):

Summary:

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The manuscript presents some novel new phenotypes generated in axolotl limbs due to Wnt signaling. This is generally the first example in which Wnt signaling has provided a gain of function in the axolotl limb model. They also present a potent way of inducing limb patterning in the dorsal axis by the addition of just beads loaded with Bmp2+Fgf8+Fgf2.

Comments on revised version:

Re-evaluation: The authors have significantly improved the manuscript and their conclusions reflect the current state of knowledge in DV patterning of tetrapod limbs. My only point of consideration is their claim of mesenchymal and epithelial expression of Wnt10b and the finding that Fgf2 and Wnt10b are lowly expressed. It is based upon the failed ISH, but this doesn't mean they aren't expressed. In interpreting the Li et al. scRNAseq dataset, conclusions depend heavily on how one analyzes and interprets it. The 7DPA sample shows a very low representation of epithelial cells compared to other time points, but this is likely a technical issue. Even the epithelial marker, Krt17, and the CT/fibroblast marker show some expression elsewhere. If other time points are included in the analysis, Wnt10b, would be interpreted as relatively highly expressed almost exclusively in the epithelium. By selecting the 7dpa timepoint, which may or may not represent the MB stage as it wasn't shown in the paper, the conclusions may be based

upon incomplete data. I don't expect the authors to do more work, but it is worth mentioning this possibility. The authors have considered and made efforts to resolve previous concerns.

We are grateful for the constructive comments. As Reviewer #1 suggested, we noted that clearer expression patterns of *Wnt10b* and *Fgf2* may be detectable in scRNA-seq analyses at other stages, and we also clarified that low-level signals of epithelial and CT/fibroblast markers outside their expected clusters may reflect technical bias in the Discussion section. In addition, we agree with the reviewer's point that our unsuccessful ISH experiments and the low abundance detected by RT-qPCR do not demonstrate absence of expression, and that conclusions from reanalyzing the Li et al. scRNA-seq dataset can depend strongly on analytical choices; therefore, while we focused on the 7 dpa sample because our RT-qPCR data suggested that *Wnt10b* and *Fgf2* may be most enriched around the MB stage (the original study refers to 7 dpa as MB), we explicitly acknowledged that analyzing a single time point—especially one with a low representation of epithelial cells—may yield incomplete or stage-biased interpretations, and that inclusion of additional datasets could reveal clearer and potentially different expression patterns in the Discussion section. We also tempered our wording regarding the inferred cellular sources to avoid over-interpretation based on the current data in the Results section.

Reviewer #2 (Public review):

Summary:

This study explores how signals from all sides of a developing limb, front/back and top/bottom, work together to guide the regrowth of a fully patterned limb in axolotls, a type of salamander known for its impressive ability to regenerate limbs. Using a model called the Accessory Limb Model (ALM), the researchers created early staged limb regenerates (called blastemas) with cells from different sides of the limb. They discovered that successful limb regrowth only happens when the blastema contains cells from both the top (dorsal) and bottom (ventral) of the limb. They also found that a key gene involved in front/back limb patterning, called Shh (Sonic hedgehog), is only turned on when cells from both the dorsal and ventral sides come into contact. The study identified two important molecules, Wnt10B and FGF2, that help activate Shh when dorsal and ventral cells interact. Finally, the authors propose a new model that explains how cells from all four sides of a limb, dorsal, ventral, anterior (front), and posterior (back), contribute at both the cellular and molecular level to rebuilding a properly structured limb during regeneration.

Strengths:

The techniques used in this study, like delicate surgeries, tissue grafting, and implanting tiny beads soaked with growth factors, are extremely difficult, and only a few research groups in the world can do them successfully. These methods are essential for answering important questions about how animals like axolotls regenerate limbs with the correct structure and orientation. To understand how cells from different sides of the limb communicate during regeneration, the researchers used a technique called in situ hybridization, which lets them see where specific genes are active in the developing limb. They clearly showed that the gene Shh, which helps pattern the front and back of the limb, only turns on when cells from both the top (dorsal) and bottom (ventral) sides are present and interacting. The team also took a broad, unbiased approach to figure out which signaling molecules are unique to dorsal and ventral limb cells. They tested these molecules individually and discovered which could substitute for actual dorsal and ventral cells, providing the same necessary signals for proper limb development. Overall, this study makes a major contribution to our understanding of how complex signals

guide limb regeneration, showing how different regions of the limb work together at both the cellular and molecular levels to rebuild a fully patterned structure.

Weaknesses:

Because the expressional analyses are performed on thin sections of regenerating tissue, in the original manuscript, they provided only a limited view of the gene expression patterns in their experiments, opening the possibility that they could be missing some expression in other regions of the blastema. Additionally, the quantification method of the expressional phenotypes in most of the experiments did not appear to be based on a rigorous methodology. The authors' inclusion of an alternate expression analysis, qRT-PCR, on the entire blastema helped validate that the authors are not missing something in the revised manuscript.

Overall, the number of replicates per sample group in the original manuscript was quite low (sometimes as low as 3), which was especially risky with challenging techniques like the ones the authors employ. The authors have improved the rigor of the experiment in the revised manuscript by increasing the number of replicates. The authors have not performed a power analysis to calculate the number of animals used in each experiment that is sufficient to identify possible statistical differences between groups. However, the authors have indicated that there was not sufficient preliminary data to appropriately make these quantifications.

Likewise, in the original manuscript, the authors used an AI-generated algorithm to quantify symmetry on the dorsal/ventral axis, and my concern was that this approach doesn't appear to account for possible biases due to tissue sectioning angles. They also seem to arbitrarily pick locations in each sample group to compare symmetry measurements. There are other methods, which include using specific muscle groups and nerve bundles as dorsal/ventral landmarks, that would more clearly show differences in symmetry. The authors have now sufficiently addressed this concern by including transverse sections of the limbs and have explained the limitations of using a landmark-based approach in their quantification strategy.

We are grateful for the careful evaluation of the technical rigor and quantification. We have benefited from the reviewer's earlier feedback, which guided revisions that improved the manuscript's rigor and presentation.

Reviewer #3 (Public review):

Summary:

After salamander limb amputation, the cross-section of the stump has two major axes: anterior-posterior and dorsal-ventral. Cells from all axial positions (anterior, posterior, dorsal, ventral) are necessary for regeneration, yet the molecular basis for this requirement has remained unknown. To address this gap, Yamamoto et al. took advantage of the ALM assay, in which defined positional identities can be combined on demand and their effects assessed through the outgrowth of an ectopic limb. They propose a compelling model in which dorsal and ventral cells communicate by secreting Wnt10b and Fgf2 ligands respectively, with this interaction inducing Shh expression in posterior cells. Shh was previously shown to induce limb outgrowth in collaboration with anterior Fgf8 (PMID: 27120163). Thus, this study completes a concept in which four secreted signals from four axial positions interact for limb patterning. Notably, this work firmly places dorsal-ventral interactions upstream of anterior-posterior, which is striking for a field that has been focussed on anterior-posterior communication. The ligands identified (Wnt10b, Fgf2) are different to those implicated in dorsal-ventral patterning in the non-regenerative mouse and chick models. The strength of this study is in the context of ALM/ectopic limb engineering. Although the authors attempt to assay the expression

of *Wnt10b* and *Fgf2* during limb regeneration after amputation, they were unable to pinpoint the precise expression domains of these genes beyond 'dorsal' and 'ventral' blastema. Given that experimental perturbations were not performed in regenerating limbs - almost exclusively under ALM conditions - this author finds the title "Dorsoventral-mediated *Shh* induction is required for axolotl limb regeneration" a little misleading.

Strengths:

(1) The ALM and use of GFP grafts for lineage tracing (Figures 1-3) take full advantage of the salamander model's unique ability to outgrow patterned limbs under defined conditions. As far as I am aware, the ALM has not been combined with precise grafts that assay 2 axial positions at once, as performed in Figure 3. The number of ALMs performed in this study deserves special mention, considering the challenging surgery involved.

(2) The authors identify that posterior *Shh* is not expressed unless both dorsal and ventral cells are present. This echoes previous work in mouse limb development models (AER/ectoderm-mesoderm interaction) but this link between axes was not known in salamanders. The authors elegantly reconstitute dorsal-ventral communication by grafting, finding that this is sufficient to trigger *Shh* expression (Figure 3 - although see also section on Weaknesses).

(3) Impressively, the authors discovered two molecules sufficient to substitute dorsal or ventral cells through electroporation into dorsal- or ventral- depleted ALMs (Figure 5). These molecules did not change the positional identity of target cells. The same group previously identified the ventral factor (*Fgf2*) to be a nerve-derived factor essential for regeneration. In Figure 6, the authors demonstrate that nerve-derived factors, including *Fgf2*, are alone sufficient to grow out ectopic limbs from a dorsal wound. Limb induction with a 3-factor cocktail without supplementing with other cells is conceptually important for regenerative engineering.

(4) The writing style and presentation of results is very clear.

Overall appraisal:

This is a logical and well-executed study that creatively uses the axolotl model to advance an important framework for understanding limb patterning. The relevance of the mechanisms to normal limb regeneration are not yet substantiated, in the opinion of this reviewer. Additionally, *Wnt10b* and *Fgf2* should be considered molecules sufficient to substitute dorsal and ventral identity (solely in terms of inducing *Shh* expression). It is not yet clear whether these molecules are truly necessary (loss of function would address this).

Comments on revisions:

Congratulations - I still find this an elegant and easy-to-read study with significant implications for the field! Linking your mechanisms to normal limb regeneration (i.e. regenerating blastema, not ALM), as well as characterising the cell populations involved, will be interesting directions for the future.

We are grateful for the constructive comments. To mitigate the concerns raised by Reviewer #3, we cited a previous study suggesting that ALM was used as the alternative experimental system for studying limb regeneration (Nacu et al., 2016, Nature, PMID: 27120163; Satoh et al., 2007, Developmental Biology, PMID: 17959163) in the Introduction section. We are confident that our ALM-based data provide a reasonable basis for understanding limb regeneration. We agree that there are important remaining questions—such as which cell populations express *Wnt10b* and *Fgf2* and how endogenous WNT10B and FGF2 signals induce *Shh* expression in

normal regeneration—which should be investigated in future studies to deepen our understanding of limb regeneration.

The following is the authors' response to the original reviews.

Recommendations for the authors:

Reviewing Editor Comments:

The authors should be commended for addressing this gap - how cues from the DV axis interact with the AP axis during limb regeneration. Overall, the concept presented in this manuscript is extremely interesting and could be of high value to the field. However, the manuscript in its current form is lacking a few important data and resolution to fully support their conclusions, and the following needs to be addressed before publication:

(1) ISH data on Wnt10b and FGF2 from various regeneration time points are essential to derive the conclusion. Preferably multiplex ISH of Wnt10b/Fgf2/Shh or at least canonical ISH on serial sections to demonstrate their expression in dermis/epidermis and order of gene expression i.e. Shh is only expressed after expression of Wnt10b/FGF2. It would certainly help if this can also be shown in regular blastema.

We are grateful for the constructive suggestion on assessing Wnt10b and Fgf2 expression during regular regeneration, and we agree that clarifying their expression patterns in regular blastemas is important for strengthening the conclusions of our study. Because we cannot currently ensure sufficient sensitivity with multiplex FISH in our laboratory—partly due to high background—, we conducted conventional ISH on serial sections of regular blastemas at several time points (Fig. S5A). However, the expression patterns of Wnt10b and Fgf2 were not clear. To complement the ISH results, we performed RT-qPCR on microdissected dorsal and ventral halves of regular blastemas at the MB stage (Fig. S5B). We found that Wnt10b and Fgf2 were expressed at significantly higher levels in the dorsal and ventral halves, respectively, compared to the opposite half. This dorsal/ventral biased expression of Wnt10b/Fgf2 is consistent with our RNA-seq data. We further quantified expression levels of Wnt10b, Fgf2, and Shh across stages (intact, EB, MB, LB, and ED) and found that Wnt10b and Fgf2 peaked at the MB stage, whereas Shh peaked at the LB stage—consistent with the editor's request regarding the order of gene expression (Fig. S5C). This temporal offset in upregulation supports our model. These results are now included in the revised manuscript (Line 294–306).

To identify the cell types expressing Wnt10b or Fgf2, we analyzed published single-cell RNA-seq data (7 dpa blastema (MB), Li et al., 2021). As a result, Fgf2 expression was observed in the mesenchymal cluster, whereas Wnt10b expression was observed in both mesenchymal and epithelial clusters (Fig. S6). However, because only a small fraction of cells expressed Wnt10b, the principal cellular source of WNT10B protein remains unclear. The apparent low abundance likely contributes to the weak ISH signals and reflects current technical limitations. In addition, Wnt10b and Fgf2 expression did not follow Lmx1b expression (Fig. S6J, K), and Wnt10b and Fgf2 themselves were not exclusive (Fig. S6L). These results are now included in the revised manuscript (Line 307–321). Together with the RT-qPCR data (Fig. S5B), these results suggest that Wnt10b and Fgf2 are not exclusively confined to purely dorsal or ventral cells at the single-cell level, even though they show dorsoventral bias when assessed in bulk tissue. These results suggest that Wnt10b/Fgf2 expression is not restricted to dorsal/ventral cells but mediated by dorsal/ventral cells, and co-existence of both signals should provide a permissive environment for Shh induction. Defining the precise spatial patterns of Wnt10b and Fgf2 in regular regeneration will therefore be an important goal for future work.

(2) Validation of the absence of gene expression via qRT PCR in the given sample will increase the rigor, as suggested by reviewers.

We thank for this important suggestion and agree that validation by qRT-PCR increases the rigor of our study. Accordingly, we performed RT-qPCR on AntBL, PostBL, DorBL, and VentBL to corroborate the ISH results. The results are now included in Fig. 2. We also verified by RT-qPCR that *Shh* expression following electroporation and the quantitative results are now provided in Fig. 5.

(3) Please increase n for experiments where necessary and mention n values in the figures.

We thank for this helpful comment and agree on the importance of providing sufficient sample sizes. Accordingly, we increased the n for the relevant experiments and have indicated the n values in the corresponding figure legends.

(4) Most comments by all three reviewers are constructive and largely focus on improving the tone and language of the manuscript, and I expect that the authors should take care of them.

We thank the reviewers for their constructive feedback on the tone and language of the manuscript. We have carefully revised the text according to each comment, and we hope these modifications have improved both clarity and readability.

In addition, in revising the manuscript we also refined the conceptual framework. Our new analysis of *Wnt10b* and *Fgf2* expression during normal regeneration suggests that these genes are not expressed in a strictly dorsal- or ventral-specific manner at the single-cell level. When these observations are considered together with (i) the RNA-seq comparison of dorsally and ventrally induced ALM blastemas, (ii) RT-qPCR of microdissected dorsal and ventral halves of regenerating blastemas, and (iii) the functional electroporation experiments, our interpretation is that *Wnt10b* and *Fgf2* act as dorsal- and ventral-mediated signals, respectively: their production is regulated by dorsal or ventral cells, and the presence of both signals is required to induce *Shh* expression. Given those, we now think our conclusion might be explained without using the confusing term, “positional cue”. Because the distinction between “positional cue” and “positional information” could be confusing as noted by the reviewers, we rewrote our manuscript without using “positional cue.”

Reviewer #1 (Recommendations for the authors):

(1) Line 61: More explanation for what a double-half limb means is needed.

We thank the reviewer for this suggestion. We have revised the manuscript (Line 73–76). Specifically, we now explain that a double-dorsal limb, for example, is a chimeric limb generated by excising the ventral half and replacing it with a dorsal half from the contralateral limb while preserving the anteroposterior orientation.

(2) Line 63-65: "Such blastemas form hypomorphic, spike-like structures or fail to regenerate entirely." This statement does not represent the breadth of work on the APDV axis in limb regeneration. The cited Bryant 1976 reference tested only double-posterior and double-anterior newt limbs, demonstrating the importance of disposition along the AP axis, not DV. Others have shown that the regeneration of double-half limbs depends upon the age of the animal and the length of time between the grafting of double-half limbs and amputation. Also, some double-dorsal or double-ventral limbs will regenerate complete AP axes with symmetrical DV duplications (Burton, Holder, and Jesani, 1986). Also, sometimes half dorsal stylopods regenerate half dorsal and half ventral, or regenerate only half ventral, suggesting there are no inductive cues across the DV axis as there are along the AP axis. Considering this is the basis of the study under question, more is needed to convince that the DV axis is necessary for the generation of the AP axis.

We thank the reviewer for this detailed and constructive comment. We acknowledge that previous studies have reported a range of outcomes for double-half limbs. For example, Burton et al. (1986) described regeneration defects in double-dorsal (DD) and double-ventral (VV) limbs, although limb patterning did occur in some cases (Burton et al., 1986, Table 1). As the reviewer notes, regenerative outcomes depend on variables such as animal age and the interval between construction of the double-half limb and amputation, sometimes called the effect of healing time (Tank and Holder, 1978). Moreover, variability has been reported not only in DD/VV limbs but also in double-anterior (AA) and double-posterior (PP) limbs (e.g., Bryant, 1976; Bryant and Baca, 1978; Burton et al., 1986). In the revised manuscript, we have therefore modified the statement to avoid over-generalization and to emphasize that regeneration can be incomplete under these conditions (Line 76–82). Importantly, in order to provide the additional evidence requested and to directly re-evaluate whether dorsal and ventral cells are required for limb patterning, we performed the ALM experiments shown in Fig. 1. The ALM system allows us to assess this question in a binary manner (regeneration vs. non-regeneration), thereby strengthening the rationale for our conclusions regarding the necessity of the APDV orientations. We also revised a sentence at the beginning of the Results section to emphasize this point (Line 139–140).

(3) Line 71: These findings suggest that specific signals from all four positional domains must be integrated for successful limb patterning, such that the absence of any one of them leads to failure." I was under the impression that half posterior limbs can grow all elements, but half anterior can only grow anterior elements.

We thank the reviewer for this helpful clarification. As summarized by Stocum, half-limb experiments show that while some digit formation can occur, limb patterning remains incomplete in both anterior-half and posterior-half limbs in some cases (Stocum, 2017). We see this point as closely related to the broader question of whether proper limb patterning requires the integration of signals from all four positional domains. As noted in our response above, our ALM experiments in Fig. 1 were designed to test this point directly, and our data support the interpretation that cells from all four orientations are necessary for correct limb patterning.

(4) Line 79-81: This is stated later in lines 98-105. I suggest expanding here or removing it here.

We thank the reviewer for this suggestion. In the original version, lines 79–81 introduced our use of the terms “positional cue” and “positional information,” and this content partially overlapped with what later appeared in lines 98–105. In the revised manuscript, we have substantially rewritten this section (Line 82–84), including the sentences corresponding to lines 79–81 in the original version, to remove the term “positional cue,” as explained in our response to the Editor’s comment (4); our revision reflects new analyses indicating that Wnt10b and Fgf2 appear not be strictly restricted to dorsal or ventral cell populations, and we now describe these factors as dorsal- or ventral-mediated signals that act across dorsoventral domains to induce Shh expression. Accordingly, we no longer maintain the original use of “positional cue” and “positional information.”

(5) Line 92 - 93: "Similarly, an ALM blastema can be induced in a position-specific manner along the limb axes. In this case, the induced ALM blastema will lack cells from the opposite side." This sentence is difficult to follow. Isn't it the same thing stated in lines 88-90?

We thank the reviewer for this comment. We revised the sentence to improve readability and to avoid redundancy with original Lines 88–90 (Line 104–106).

(6) Line 107: I think the appropriate reference is McCusker et al., 2014 (Position-specific induction of ectopic limbs in non-regenerating blastemas on axolotl forelimbs), although

Vieira et al., 2019 can be included here. In addition, Ludolph et al 1990 should be cited.

We thank the reviewer for this suggestion. We have added McCusker et al. (2014) and Ludolph et al. (1990) as references in the revised manuscript (Line 120–121).

(7) Line 107-109: A missing point is how the ventral information is established in the amniote limb. From what I remember, it is the expression of Engrailed 1, which inhibits the ventral expression of Wnt7a, and hence Lmx1b. This would suggest that there is no secreted ventral cue. This is a relatively large omission in the manuscript.

We thank the reviewer for this comment. We agree that ventral fate in amniotes is specified by En1 in the ventral ectoderm, which represses Wnt7a and thereby prevents induction of Lmx1b; accordingly, a secreted ventral morphogen analogous to dorsal Wnt7a has not been established. We added this point to the revised Introduction (Line 61–64).

By contrast, in axolotl limb regeneration, our previous work on Lmx1b expression suggests that DV identities reflect the original positional identity rather than being re-specified during regeneration (Yamamoto et al., 2022). Within this framework, our original use of the term “ventral positional cue” does not imply a ventral patterning morphogen in the amniote sense; rather, it denotes downstream signals induced by cells bearing ventral identity that are required for the blastema to form a patterned limb. This interpretation is consistent with classic studies on double-half chimeras and ectopic contacts between opposite regions (Iten & Bryant, 1975; Bryant & Iten, 1976; Maden, 1980; Stocum, 1982) as well as with our ALM data (Fig. 1). For this reason, we intentionally used the term “positional cues” to refer to signals provided by cells bearing ventral identity, which can be considered separable from the DV patterning mechanism itself, in the original text. As explained in our response to the Editor’s comment (4), we describe these signals as “signals mediated by dorsal/ventral cells,” rather than “positional cues” in the revised manuscript.

The necessity of dorsal- and ventral-mediated signals is supported by classic studies on the double-half experiment. In the non-regenerating cases, structural patterns along the anteroposterior axis appear to be lost even though both anterior and posterior cells should, in principle, be present in a blastema induced from a double-dorsal or double-ventral limbs. In limb development of amniotes, Wnt7a/Lmx1b or En-1 mutants show that limbs can exhibit anteroposterior patterning even when tissues are dorsalized or ventralized—that is, in the relative absence of ventral or dorsal cells, respectively (Riddle et al., 1995; Chen et al., 1998; Loomis et al., 1996). Taken together, axolotl limb regeneration, in which the presence of both dorsal and ventral cells plays a role in anteroposterior patterning, should differ from other model organisms. It is reasonable to predict the dorsal- and ventral-mediated signals in axolotl limb regeneration. We included this point in the revised manuscript (Line 82–89). However, there is no evidence that these signals are secreted molecules. For this reason, we have carefully used the term “dorsal-/ventral-mediated signals” in the Introduction without implying secretion.

(8) Introduction - In general, the argument is a bit misleading. It is written as if it is known that a ventral cue is necessary, but the evidence from other animal models is lacking, from what I know. I may be wrong, but further argument would strengthen the reasoning for the study.

We thank the reviewer for this thoughtful comment. We agree that it should not read as if it is known that a ventral cue is necessary. In the revised Introduction, we have addressed this in several ways. First, as described in our response to comment (7), we now explicitly note that in amniote limb development ventral identity is specified by En1-mediated repression of Wnt7a, and that a secreted ventral morphogen equivalent to dorsal Wnt7a has not been established. Second, we removed the term “positional cue” and no longer present “ventral positional cue” as a defined entity. Instead, we use mechanistic phrasing such as “signals

mediated by ventral cells” and “signals mediated by dorsal cells,” which does not assume that such signals are secreted morphogens or universally conserved. Third, we have reframed the role of dorsal- and ventral-mediated signals as a working hypothesis specific to axolotl limb regeneration, rather than as a general conclusion across model systems.

| (9) Line 129: Remove “As mentioned before”.

We thank the reviewer for this suggestion. We have removed the phrase “As mentioned before” in the revised manuscript (Line 143).

| (10) Figure 1: Are *Lmx1*, *Fgf8*, and *Shh* mutually exclusive? Multiplexed FISH would provide this information, and is a relatively important question considering the strong claims in the study.

We thank the reviewer for raising this important point. As noted in our response to the editor’s comment, we cannot currently ensure sufficiently high detection sensitivity with multiplex FISH in our laboratory. However, based on previous reports (Nacu et al., 2016), *Fgf8* and *Shh* should be mutually exclusive. In contrast, with respect to *Lmx1b*, our analysis suggests that its expression is not mutually exclusive with either *Fgf8* or *Shh*, at least their expression domains. To confirm this, we analyzed the published scRNA-seq data and the results were added to the supplemental figure 6. *Fgf8* and *Shh* were expressed in both *Lmx1b*-positive and *Lmx1b*-negative cells (Fig. S6H, I), but *Fgf8* and *Shh* themselves were mutually exclusive (Fig. S6M). This point is now included in the revised manuscript (Line 314–317).

| (11) Results section and Figure 2: More evidence is needed for the lack of *Shh* expression ISH in tissue sections. Demonstrating the absence of something needs some qPCR or other validation to make such a claim.

We thank the reviewer for this suggestion. We performed qRT-PCR on ALM blastemas to complement the ISH data (Fig. 2).

| (12) Line 179: I think they are likely leucistic d/d animals and not wild-type animals based upon the images.

We thank the reviewer for this observation. In the revised manuscript, we have corrected the description to “leucistic animals” (Line 194).

| (13) Line 183-186: I’m a bit confused about this interpretation. If *Shh* turns on in just a posterior blastema, wouldn’t it turn on in a grafted posterior tissue into a dorsal or ventral region? Isn’t this independent of environment, meaning *Shh* turns on if the cells are posterior, regardless of environment?

Our interpretation is that only posterior-derived cells possess the competency to express *Shh*. In other words, whether a cell is capable of expressing *Shh* depends on its original positional identity (Iwata et al., 2020), but whether it actually expresses *Shh* depends on the environment in which the cell is placed. The results of Fig. 3E and G indicate that *Shh* activation is dependent on environment and that the posterior identity is not sufficient to activate *Shh* expression. We have revised the manuscript to emphasize this distinction more clearly (Line 198–203).

| (14) Figure 4: Do the limbs have an elbow, or is it just a hand?

We thank the reviewer for this thoughtful question. From the appearance, an elbow-like structure can occasionally be seen; however, we did not examine the skeletal pattern in detail because all regenerated limbs used for this analysis were sectioned for the purpose of symmetry evaluation, and we therefore cannot state this conclusively. While this is indeed an important point, analyzing proximodistal patterning would require a very large number of

additional experiments, which falls outside the main focus of the present study. For this reason, and also to minimize animal use in accordance with ethical considerations, we did not pursue further experiments here. In response to this point, we have added a description of the skeletal morphology of ectopic limbs induced by BMP2+FGF2+FGF8 bead implantation (Fig. 6). In these experiments, multiple ectopic limbs were induced along the same host limb. In most cases, these ectopic limbs did not show fusion with the proximal host skeleton, similar to standard ALM-induced limbs, although in one case we observed fusion at the stylopod level. We now note this observation in the revised manuscript (Line 347–354).

We regard the relationship between APDV positional information and proximodistal patterning as an important subject for future investigation.

(15) Line 203 - 237: I appreciate the symmetry score to estimate the DV axis. Are there landmarks that would better suggest a double-dorsal or double-ventral phenotype, like was done in the original double-half limb papers?

We thank the reviewer for this thoughtful comment. In most cases, the limbs induced by the ALM exhibit abnormal and highly variable morphologies compared to normal limbs, making it difficult to apply consistent morphological landmarks as used in the original double-half limb studies. For this reason, we focused our analysis on “morphological symmetry” as a quantitative measure of DV axis patterning, and we have added this explanation to the manuscript (Line 232–235). Additionally, we provided transverse sections along the proximodistal axis as supplemental figures (Figs. S2 and S4). In addition to reporting the symmetry score, we have explicitly stated in the text that symmetry was also assessed by visual inspection of these sections.

(16) Line 245-247: The experiment was done using bulk sequencing, so both the epithelium and mesenchyme were included in the sample. The posterior (Shh) and anterior (Fgf8) patterning cues are mesenchymally expressed. In amniotes, the dorsal cue has been thought to be Wnt7a from the epithelium. Can ISH, FISH, or previous scRNAseq data be used to identify genes expressed in the mesenchyme versus epithelium? This is very important if the authors want to make the claim for defining "The molecular basis of the dorsal and ventral positional cues" as was stated by the authors.

We thank the reviewer for highlighting this important point. As the reviewer notes, our bulk RNA-seq data do not distinguish between epithelial and mesenchymal expression domains. As noted in our response to the editor's comment, we performed ISH and qPCR on regular blastemas. However, these approaches did not provide definitive information regarding the specific cell types expressing Wnt10b and Fgf2. To complement this, we re-analyzed publicly available single-cell RNA-seq data (from Li et al., 2021). As a results, Fgf2 was expressed mainly by the mesenchymal cells, and Wnt10b expression was observed in both mesenchymal and epithelial cells. These results are now included in the revised manuscript (Line 294–321) and in supplemental figures (Fig. S6, S7).

(17) Was engrailed 1, Lmx1b, or Wnt7a differentially expressed along the DV axis, suggesting similar signaling between? Are these expressed in mesenchyme? Previous work suggests Wnt7a is expressed throughout the mesenchyme, but publicly available scRNAseq suggests that it is expressed in the epithelium.

We thank the reviewer for this important comment. As noted, the reported expression patterns of DV-related genes are not consistent across studies, which likely reflects the technical difficulty of detecting these genes with high sensitivity. In our own experiments, expression of DV markers other than Lmx1b has been very weak or unclear by ISH. Whether these genes are expressed in the epithelium or mesenchyme also appears to vary depending on the detection method used. In our RNA-seq dataset, Wnt7a expression was detected at very low levels and showed no significant difference along the DV axis, while En1 expression

was nearly absent. We have clarified these results in the revised manuscript (Line 437–441). Our reanalysis of the published scRNA-seq likewise detected Wnt7a in only a very small fraction of cells. Accordingly, we consider it premature to reach a definitive conclusion—such as whether Wnt7a is broadly mesenchymal or restricted to epithelium—as suggested in prior reports. We also note that whether Wnt7a is epithelial or mesenchymal does not affect the conclusions or arguments of the present study. Although the roles of Wnt7a and En1 in axolotl DV patterning are certainly important, we feel that drawing a definitive conclusion on this issue lies beyond the scope of the present study, and we have therefore limited our description to a straightforward presentation of the data.

(18) Line 247-249: The sentence suggests that all the ligands were tried. This should be included in the supplemental data.

We thank the reviewer for this clarification. In fact, we tested only Wnt4, Wnt10b, Fgf2, Fgf7, and Tgfb2, and all of these results are presented in the figures. To avoid misunderstanding, we have revised the text to explicitly state that our analysis focused on these five genes (Line 272–274).

(19) Line 249: An $n = 3$ seems low and qPCR would be a more sensitive means of measuring gene induction compared to ISH. The ISH would confirm the qPCR results. Figure 5C is also not the most convincing image of Shh induction without support from a secondary method.

We have increased the sample size for these experiments (Line 277–280). In addition, to complement the ISH results, we confirmed Shh induction by qPCR following electroporation of Wnt10b and Fgf2 (Fig. 5D, E). In addition, because Shh signal in the Wnt10b-electroporated VentBL images was particularly weak and difficult to discern, we replaced that panel with a representative example in which Shh signal is more clearly visible. These data are now included in the revised manuscript (Line 280–282).

(20) Line 253: It is confusing why Wnt10b, but not Wnt4 would work? As far as I know, both are canonical Wnt ligands. Was Wnt7a identified as expressed in the RNAseq, but not dorsally localized? Would electroporation of Wnt7a do the same thing as Wnt10b and hence have the same dorsalizing patterning mechanisms as amniotes?

We thank the reviewer for raising this challenging but important question. Wnt10b was identified directly from our bulk RNA-seq analysis, as was Wnt4. The difference in the ability of Wnt10b and Wnt4 to induce Shh expression in VentBL may reflect differences in how these ligands activate downstream WNT signaling programs. WNT10B is a potent activator of the canonical WNT/ β -catenin pathway (Bennett et al., 2005), although WNT10B has also been reported to trigger a β -catenin-independent pathway (Lin et al., 2021). By contrast, WNT4 can signal through both canonical and non-canonical (β -catenin-independent) pathways, and the balance between these outputs is known to depend on cellular context (Li et al., 2013; Li et al., 2019). Consistent with a requirement for canonical WNT signaling, we found that pharmacological activation of canonical WNT signaling with BIO (a GSK3 inhibitor) was also sufficient to induce Shh expression in VentBL. However, despite this, it is still unclear why Wnt10b, but not Wnt4, was able to induce Shh under our experimental conditions. One possible explanation is that different WNT ligands can engage the same receptors (e.g., Frizzled/LRP6) yet can drive distinct downstream transcriptional programs (This may depend on the state of the responding cells, as Voss et al. predicted), resulting in ligand-specific outputs (Voss et al., 2025). This point is now included in the revised discussion section (Line 402–412). At present, we cannot distinguish between these possibilities experimentally, and we therefore refrain from making a stronger mechanistic claim.

With respect to Wnt7a, we detected Wnt7a expression at very low levels, and without a clear dorsoventral bias, in our RNA-seq analysis of ALM blastemas (we describe this point in Line

437–440). This is consistent with previous work suggesting that axolotl Wnt7a is not restricted to the dorsal region in regeneration. Because of this low and unbiased expression, and because our data already implicated Wnt10b as a dorsal-mediated signal that can act across dorsoventral domains to permit Shh induction, we did not prioritize Wnt7a electroporation in the present study. We therefore cannot conclude whether Wnt7a would behave similarly to Wnt10b in this context.

Importantly, these uncertainties about ligand-specific mechanisms do not alter our main conclusion. Our data support the idea that a dorsal-mediated WNT signal (represented here by WNT10B and canonical WNT activation) and a ventral-mediated FGF signal (FGF2) must act together to permit Shh induction, and that the coexistence of these dorsal- and ventral-mediated signals is required for patterned limb formation in axolotl limb regeneration.

(21) Is canonical Wnt signaling induced after electroporation of Wnt10b or Wnt4? qPCR of Lef1 and axin is the most common way of showing this.

We thank the reviewer for this helpful suggestion. In addition to examining Shh expression, we also assessed canonical WNT signaling by qPCR analysis of Axin2 and Lef1 following Wnt10b electroporation. The data is now included in Fig. 5.

(22) Line 255-256: qPCR was presented for Figure 5D, but ISH was used for everything else. Is there a technical reason that just qPCR was used for the bead experiments?

We thank the reviewer for this helpful comment. In the original submission, our goal was to test whether treatment with commercial FGF2 protein or BIO could reproduce the results obtained by electroporation. In the revised manuscript, to avoid confusion between distinct experimental aims, we removed the FGF2–bead data from this section and instead used RT-qPCR to quantitatively corroborate Shh induction after electroporation (Fig. 5D–E). RT-qPCR provided a sensitive, whole-blastema readout and allowed a paired design (left limb: factor; right limb: GFP control) that increased statistical power while minimizing animal use. To address the reviewer’s point more directly, we additionally performed ISH for the BIO treatment and now include those results in Supplementary Figure 3 (Line 287–288).

(23) Line 261-263: The authors did not show where Wnt10b or Fgf2 is expressed in the limb as claimed. The RNAseq was bulk, so ISH of these genes is needed to make this claim. Where are Wnt10b and Fgf2 expressed in the amputated limb? Do they show a dorsal (Wnt10b) and ventral (Fgf2) expression pattern?

We thank the reviewer for raising this important point. As noted in our response to the editor’s comment, we performed ISH on serial sections of regular blastemas at several time points (Fig. S5A). However, the expression patterns of Wnt10b and Fgf2 along the dorsoventral axis were not clear. To complement the ISH results, we performed RT-qPCR on microdissected dorsal and ventral halves of regular blastemas at the MB stage (Fig. S5B). We found that Wnt10b and Fgf2 were expressed at significantly higher levels in the dorsal and ventral halves, respectively, compared to the opposite half. This dorsal/ventral biased expression of Wnt10b/Fgf2 is consistent with our RNA-seq data. To identify the cell types expressing Wnt10b or Fgf2, we analyzed published single-cell RNA-seq data (7 dpa blastema (MB), Li et al., 2021). As a result, Fgf2 expression was observed in the mesenchymal cluster, whereas Wnt10b expression was observed in both mesenchymal and epithelial clusters (Fig. S6). However, because only a small fraction of cells expressed Wnt10b, the principal cellular source of WNT10B protein remains unclear. The apparent low abundance likely contributes to the weak ISH signals and reflects current technical limitations. In addition, Wnt10b and Fgf2 expression did not follow Lmx1b expression (Fig. S6J, K), and Wnt10b and Fgf2 themselves were not exclusive (Fig. S6L). Together with the RT-qPCR data (Fig. S5B), these results suggest that Wnt10b and Fgf2 are not exclusively confined to purely dorsal or ventral cells at the single-cell level, even though they show dorsoventral bias when assessed in bulk

tissue, suggesting that Wnt10b/Fgf2 expression is not dorsal-/ventral-specific but mediated by dorsal/ventral cells. Defining the precise spatial patterns of Wnt10b and Fgf2 in regular regeneration will therefore be an important goal for future work. These points are now included in the revised manuscript (Line 485–501).

(24) Line 266-288: The formation of multiple limbs is impressive. Do these new limbs correspond to the PD location they are generated?

We thank the reviewer for this interesting question. Interestingly, from our observations, there does appear to be a tendency for the induced limbs to vary in length depending on their PD location. The skeletal patterns of the induced multiple limbs are now included in Fig. 6. However, as noted earlier, the supernumerary limbs exhibit highly variable morphologies, and a rigorous analysis of PD correlation would require a large number of induced limbs. Since this lies outside the main focus of the present study, we have not pursued this point further in the manuscript.

(25) Line 288: The minimal requirement for claiming the molecular basis for DV signaling was identified is to ISH or multiplexed FISH for Wnt10b and Fgf2 in amputated limb blastemas to show they are expressed in the mesenchyme or epithelium and are dorsally and ventrally expressed, respectively. In addition, the current understanding of DV patterning through Wnt7a, Lmx1b, and En1 shown not to be important in this model.

We thank the reviewer for this comment and fully agree with the point raised. We would like to clarify that we are not claiming to have identified the molecular basis of DV patterning. As the reviewer notes, molecules such as Lmx1b, Wnt7a, and En1 are well identified in other animal models as key regulators of DV positional identity. There is no doubt that these molecules play central roles in DV patterning. However, in axolotl limb regeneration, clear DV-specific expression has not been demonstrated for these genes except for Lmx1b. Therefore, further studies will be required to elucidate the molecular basis of DV patterning in axolotls.

Our focus here is more limited: we aim to identify the molecular basis for the mechanisms in which positional domain-mediated signals (FGF8, SHH, WNT10B, and FGF2) regulate the limb patterning process, rather than the molecular basis of DV patterning. In fact, our results on Wnt10b and Fgf2 suggest that these genes did not affect dorsoventral identities.

We recognize that this distinction was not sufficiently clear in the original text, and we have revised the manuscript to describe DV patterning mechanisms in other animals and clarify that the dorsal- and ventral-mediated signals are distinct from DV patterning (Line 444–450). At least, we avoid claiming that the molecular basis for DV signaling was identified.

(26) Line 335: References are needed for this statement. From what I found, Wnt4 can be canonical or non-canonical.

We thank the reviewer for this helpful comment. We have revised the manuscript (Line 404–407). We added these citations at the relevant location and adjusted nearby wording to avoid implying pathway exclusivity, in alignment with our response to comment (20).

(27) Line 337-338: The authors cannot claim "that canonical, but not non-canonical, WNT signaling contributes to Shh induction" as this was not thoroughly tested is based upon the negative result that Wnt4 electroporation did not induce Shh expression.

We thank the reviewer for this important clarification. We agree that our data do not allow us to conclude that non-canonical WNT signaling in general does not contribute to Shh induction. Accordingly, we have removed the phrase “but not non-canonical” and revised the text to emphasize that, within the scope of our experiments, Shh induction was not observed following Wnt4 electroporation, whereas it was observed with Wnt10b.

(28) Line 345: In order to claim "WNT10B via the canonical WNT pathway...appears to regulate Shh expression" needs at least qPCR to show WNT10B induces canonical signaling.

We thank the reviewer for this comment. As noted in our response to comment (21), we also assessed canonical WNT signaling by qPCR analysis of Axin2 and Lef1 following Wnt10b electroporation (Line 282–285).

(29) Lines 361-372: A few studies have been performed on DV patterning of the mouse digit regeneration in regards to Lmx1b and En1. It may be good to discuss how the current study aligns with these findings.

We appreciate the reviewer's suggestion. As the reviewer refers, several studies have been performed on dorsoventral (DV) patterning in mouse digit tip regeneration in relation to Lmx1b and En1 (e.g., Johnson et al., 2022; Castilla-Ibeas et al., 2023). In the present study, however, our main conclusion is different in the scope of studies on mouse digit tip regeneration. We show that, in the axolotl, pre-existing dorsal and ventral identities (as reflected by dorsally derived and ventrally derived cells in the ALM blastema) are required together to induce Shh expression, and that this Shh induction in turn supports anteroposterior interaction at the limb level. This mechanism—dorsal-mediated and ventral-mediated signals acting in combination to permit Shh expression—does not have a clear direct counterpart in the mouse digit tip literature. Moreover, even with respect to Lmx1b, the two systems behave differently. In mouse digit tip regeneration, loss of Lmx1b during regeneration does not grossly affect DV morphology of the regenerate (Johnson et al., 2022). By contrast, in our axolotl ALM system, the presence or absence of Lmx1b-positive dorsal tissue correlates with the final dorsoventral organization of the induced limb-like structures (e.g., production of double-dorsal or double-ventral symmetric structures in the absence of appropriate dorsoventral contact). Thus, the role of dorsoventral identity in our model is directly tied to patterned limb outgrowth at the whole-limb scale, whereas in the mouse digit tip it has been reported primarily in the context of digit tip regrowth and bone regeneration competence, not robust DV repatterning (Johnson et al., 2022).

For these reasons, we believe that an extended discussion of mouse digit tip regeneration would risk implying a mechanistic equivalence between axolotl limb regeneration and mouse digit tip regeneration that is not supported by current data. Because the regenerative contexts differ, and because Lmx1b does not appear to re-establish DV patterning in the mouse regenerates (Johnson et al., 2022), we have chosen not to include an explicit discussion of mouse digit tip regeneration in the main text.

(30) Line 408-433: Although I appreciate generating a model, this section takes some liberties to tell a narrative that is not entirely supported by previous literature or this study. For example, lines 415-416 state "Wnt10b and Fgf2 are expressed at higher levels in dorsal and the ventral blastemal cells, respectively" which were not shown in the study or other studies.

We thank the reviewer for this important comment. We agree that the original model based on RNA-seq data overstated the evidence. To address this point experimentally, we examined Wnt10b and Fgf2 expression in regular blastemas (Supplemental Figure 5 and 6). Accordingly, our model is now framed as an inductive mechanism for Shh expression—supported by results in ALM (WNT10B in VentBL; FGF2 in DorBL) and by DV-biased expression. Concretely, the sentence previously paraphrased as “Wnt10b and Fgf2 are expressed at higher levels in dorsal and ventral blastemal cells, respectively” has been replaced with wording that (i) avoids single-cell DV specificity and (ii) emphasizes dorsal-/ventral-mediated regulation and the requirement for both signals to allow Shh induction (Line 510–511).

Reviewer #2 (Recommendations for the authors):

(1) Introduction:

The authors' definitions of positional cues vs positional information are a little hard to follow, and do not appear to be completely accurate. From my understanding of what the authors explain, "positional information" is defined as a signal that generates positional identities in the regenerating tissue. This is a somewhat different definition than what I previously understood, which is the intrinsic (likely epigenetic) cellular identity associated with specific positional coordinates. On the other hand, the authors define "positional cues" as signals that help organize the cells according to the different axes, but don't actually generate positional identities in the regenerating cells. The authors provide two examples: Wnt7a as an example of positional information, and FGF8 as a positional cue. I think that according to the authors definitions, FGF8 (and probably Shh) are bone fide positional cues, since both signals work together to organize the regenerating limb cells - yet do not generate positional identities, because ectopic limbs formed from blastemas where these pathways have been activated do not regenerate (Nacu et al 2016). However, I am not sure Wnt7a constitutes an example of a "positional information" signal, since as far as I know, it has not been shown to generate stable dorsal limb identities (that remain after the signal has stopped) - at least yet. If it has, the authors should cite the paper that showed this. I think that some sort of diagram to help define these visually will be really helpful, especially to people who do not study regenerative patterning.

We thank the reviewer for this thoughtful comment. We now agree with the reviewer that our use of “positional cue” and “positional information” may have been confusing. In the revision—and as noted in our response to the Editor’s comment (4)—we have removed the term “positional cue” and no longer attempt to contrast it with “positional information.” Instead, we adopt phrasing that reflects our data and hypothesis: during limb patterning, dorsal-mediated signals act on ventral cells and ventral-mediated signals act on dorsal cells to induce Shh expression. This wording avoids implying that these signals specify dorsoventral identity.

Regarding WNT7A, we agree it has not been shown to generate a stable dorsal identity after signal withdrawal. In the revised Introduction we therefore describe WNT7A in amniote limb development as an extracellular regulator that induces Lmx1b in dorsal mesenchyme (with En1 repressing Wnt7a ventrally), rather than labeling it as “positional information” in a strict, identity-imprinting sense. We highlight this contrast because, in our axolotl experiments, WNT10B and FGF2 did not alter Lmx1b expression or dorsal-ventral limb characteristics when overexpressed, consistent with the idea that they act downstream of DV identity to enable Shh induction, not to establish DV identity.

(2) Results:

It would be helpful if the number of replicates per sample group were reported in the figure legends.

We thank the reviewer for this suggestion. In accordance with the comment, we have added the number of replicates (n) for each sample group in the figure legends.

Figure 2 shows ISH for A/P and D/V transcripts in different-positioned blastemas without tissue grafts. The images show interesting patterns, including the lack of Shh expression in all blastemas except in posterior-located blastemas, and localization of the dorsal transcript (Lmx1b) to the dorsal half of A or P located blastemas. My only concern about this data is that the expression patterns are described in only a small part of the ectopic blastema (how representative is it?) and the diagrams infer that these expression patterns are reflective of the entire blastema, which can't be determined by the limited field of view. It is okay if the expression patterns are not present in the entire blastema -

in fact, that might be an important observation in terms of who is generating (and might be receiving) these signals.

We thank the reviewer for this insightful comment. Because *Fgf8* and *Shh* expression was detectable only in a limited subset of cells, the original submission included only high-magnification images. In response to the reviewer's valid concern about representativeness, we have now added low-magnification overviews of the entire blastema as a supplemental figure (Fig. S1) and clarified in the figure legend that these expression patterns can be focal rather than pan-blastemal (Line 795–796).

In Figure 3, they look at all of these expression patterns in the grafted blastemas, showing that Shh expression is only visible when both D and V cells are present in the blastema. My only concern about this data is that the number of replicates is very low (some groups having only an N=3), and it is unclear how many sections the authors visualized for each replicate. This is especially important for the sample groups where they report no Shh expression -I agree that it is not observable in the single example sections they provide, but it is uncertain what is happening in other regions of the blastema.

We thank the reviewer for this important comment. To increase the reliability of the results, we have increased the number of biological replicates in groups where n was previously low. For all samples, we collected serial sections spanning the entire blastema. For blastemas in which *Shh* expression was observed, we present representative sections showing the signal. For blastemas without detectable *Shh* expression, we selected a section from the central region that contains GFP-positive cells for the Figure. To make these points explicit, we have added the following clarification to the Fig. 3 legend (Line 811–815).

Figure 4: Shh overexpression in A/P/D/V blastemas - expression induces ectopic limbs in A/D/V locations. They analyzed the symmetry of these regenerates (assuming that D and V located blastemas will exhibit D/V symmetry because they only contain cells from one side of that axis. I am a little concerned about how the symmetry assay is performed, since oblique sections through the digits could look asymmetric, while they are actually symmetric. It is also unclear how the angle of the boxes that the symmetry scores were based on was decided - I imagine that the score would change depending on the angle. It also appears that the authors picked different digits to perform this analysis on the different sample groups. I also admit that the logic of classification scheme that the authors used AI to perform their symmetry scoring analysis (both in Figures 4 and 5) is elusive to me. I think it would have been more informative if the authors leveraged the structural landmarks, like the localization of specific muscle groups. (If this experiment were performed in WT animals, the authors could have used pigment cell localization)... or generate more proximal sections to look at landmarks in the zeugopod.

We thank the reviewer for these detailed comments regarding the symmetry analysis. Because reliance on a computed symmetry score alone could raise the concerns noted by the reviewer, we now provide transverse sections along the proximodistal axis as supplemental figures (Figs. S2 and S4). These include levels corresponding to the distal end of the zeugopod and the proximal end of the autopod. In addition to reporting the symmetry score, we have explicitly stated in the text that symmetry was also assessed by visual inspection of these sections.

As also noted in our response to Reviewer #1 (comment 15), ALM-induced limbs frequently exhibit abnormal and highly variable morphologies, which makes it difficult to use consistent anatomical landmarks such as particular digits or muscle groups. For this reason, we focused our analysis on morphological symmetry rather than landmark-based metrics, and we emphasize this rationale in the revised text (Line 232–235).

Regarding the use of bounding boxes, this procedure was chosen to minimize the effects of curvature or fixation-induced distortion. For each section, the box angle was adjusted so that the outer contour (epidermal surface) was aligned symmetrically; this procedure was applied uniformly across all conditions to avoid bias. We analyzed multiple biological replicates in each group, which helps mitigate potential artifacts due to oblique sectioning. To further reduce bias, we increased the number of fields included in the analysis to $n = 24$ per group in the revised version.

In addition, staining intensity varied among samples, such that a region identified as “muscle” in one sample could be assigned differently in another if classification were based solely on color. To avoid this problem, we used a machine-learning classifier trained separately for each sample, allowing us to group the same tissues consistently within that sample irrespective of intensity differences. In the context of ALM-induced limbs, where stable anatomical landmarks are not available, we consider this strategy the most appropriate. We have added this rationale to the revised manuscript for clarity (Line 239–247).

Figure 5: The number of replicates in sample groups is relatively low and is quite variable between groups (ranging between 3 and 7 replicates). Zoom in to visualize Shh expression is small relative to the blastema, and it is difficult to discern why the authors positioned the window where they did, and how they maintained consistency among their different sample groups. In the examples of positive Shh expression - the signal is low and hard to see. Validating these expression patterns using some sort of quantitative transcriptional assay (like qRT-PCR) would increase the rigor of this experiment ... especially given that they will be able to analyze gene expression in the entire blastema as opposed to sections that might not capture localized expression.

We thank the reviewer for this important comment. To increase the rigor of these experiments, we have increased the number of biological replicates in groups where n was previously low. In addition, because *Shh* signal in the *Wnt10b*-electroporated VentBL images was particularly weak and difficult to discern, we replaced that panel with a representative example in which *Shh* signal is more clearly visible. We also validated the *Shh* expression for *Wnt10b*-electroporated VentBL and *Fgf2*-electroporated DorBL by RT-qPCR, which assesses gene expression across the entire blastema. These results are now included in Fig. 5 and Line 280–282. Finally, we clarified in the figure legend how the “window” for imaging was chosen: for samples with detectable *Shh* expression, the window was placed in the region where the signal was observed; for conditions without detectable *Shh* expression, the window was positioned in a comparable region containing GFP-positive cells (Line 836–839). These revisions are included in the revised manuscript.

Figure 6: They treat dorsal and ventral wounds with gelatin beads soaked in a combination of BMP2+FGF8 (nerve factors) and FGF2 proposed ventral factor). Remarkably, they observe ectopic limb expression in only dorsal wounds, further supporting the idea that FGF2 provides the “ventral” signal. They show examples of this impressive phenotype on limbs with multiple ectopic structures that formed along the Pr/Di axis. Including images of tubulin staining (as they have in Figures 1 and 2) to ensure that the blastemas (or final regenerates) are devoid of nerves. The authors' whole-mount skeletal staining which shows fusion of the ectopic humerus with the host humerus, is a phenotype associated with deep wounding, which could provide an opportunity for more cellular contribution from different limb axes.

We thank the reviewer for these constructive comments. As noted in the prior study, when beads are used to induce blastemas without surgical nerve orientation, fine nerve ingrowth can still occur (Makanae et al., 2014), and the induced blastemas are not completely devoid of nerves. While it is still uncertain whether these recruited nerves are functional after

blastema induction, it is an important point, and we added sentences about this in the revised manuscript (Line 341–345).

Regarding the skeletal phenotype, despite careful implantation to avoid injuring deep tissues, bead-induced ectopic limbs on the dorsal side occasionally displayed fusion of the stylopod with the host humerus—a phenotype associated with deep wounding, as the reviewer notes. This observation suggests that contributions from a broader cellular population cannot be excluded. However, because fusion was observed in only 1 of 16 induced limbs analyzed, and because ectopic limbs induced at the forearm (zeugopod) level did not exhibit such fusion ($n=1/6$ for stylopod-level inductions; $n=0/10$ for zeugopod-level inductions), we believe that our main conclusion remains valid. Because fusion is not a typical outcome, we now present representative non-fusion cases—including zeugopod-origin examples—in the figure (Fig. 6L1, L2), and we report the fusion incidence explicitly in the text (Line 350–354). We also note in the revised manuscript that stylopod fusion can occur in a minority of cases (Line 347–349).

Figure 7 nicely summarizes their findings and model for patterning.

We thank the reviewer for this positive comment.

The table is cut off in the PDF, so it cannot be evaluated at this time.

In our copy of the PDF, the table appears in full, so this may have been a formatting issue. We have carefully checked the file and ensured that the table is completely included in the revised submission.

There is a supplemental figure that doesn't seem to be referenced in the text.

The supplemental figure (Fig. S1 of the original manuscript) is referenced in the text, but it may have been overlooked. To improve clarity, we have expanded the description in the manuscript so that the supplemental figure is more clearly referenced (Line 285–291).

(3) Materials and Methods:

No power analysis was performed to calculate sample group sizes. The authors have used these experimental techniques in the past and could have easily used past data to inform these calculations.

We thank the reviewer for this important comment. We did not include a power analysis in the manuscript because this was the first time we compared *Shh* and other gene expression levels among ALM blastemas of different positional origins using RT-qPCR in our experimental system. As we did not have prior knowledge of the expected variability under these specific conditions, it was difficult to predetermine appropriate sample sizes.

Reviewer #3 (Recommendations for the authors):

General:

*Congratulations - I found this an elegant and easy-to-read study with significant implications for the field! If possible, I would urge you to consider adding some more characterisation of *Wnt10b* and *Fgf2*- which cell types are they expressed in? If you can link your mechanisms to normal limb regeneration too (i.e., regenerating blastema, not ALM), this would significantly elevate the interest in your study.*

We sincerely thank the reviewer for these encouraging comments. As also noted in our response to the editor's comment, we have analyzed the expression patterns of *Wnt10b* and *Fgf2* in regular blastemas (Line 294–306). Although clear specific expression patterns along dorsoventral axis were not detected by ISH, likely due to technical limitations of sensitivity,

RT-qPCR revealed significantly higher expression levels of Wnt10b in the dorsal half and Fgf2 in the ventral half of a regular blastema (Fig. S5). In addition, we analyzed published single-cell RNA-seq data (7 dpa blastema, Li et al., 2021) (Line 307–321). As a result, Fgf2 expression was observed in the mesenchymal clusters, whereas Wnt10b expression was observed in both mesenchymal and epithelial clusters (Fig. S6). However, because only a small fraction of cells expressed Wnt10b, the principal cellular source of WNT10B protein remains unclear. Therefore, defining the precise spatial patterns of Wnt10b and Fgf2 in regular regeneration will be an important goal for future work.

Data availability:

I assume that the RNA-sequencing data will be deposited at a public repository.

RNA-seq FASTQ files have been deposited in the DNA Data Bank of Japan (DDBJ; <https://www.ddbj.nig.ac.jp/>) under BioProject accession PRJDB38065. We have added a Data availability section to the revised manuscript.

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